

Research Article

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Nests of limit cycles in quadratic systems

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Abstract: We give a proof of the distribution property of limit cycles in so-called quadratic systems. We prove that the possible limit cycle distributions are either $(n, 0)$ or $(n, 1)$ (where $n \in \{0\} \cup \mathbb{N}$). The aim of this article is to simplify and fill gaps in the original proof by Zhang (*On the distribution and number of limit cycles for quadratic systems with two foci*, Qual. Theory Dyn. Sys. **3** (2002), 437–463). The sixteenth Hilbert problem asks for an upper bound for the number of limit cycles in polynomial systems $H(n)$, where n is the maximum degree of the polynomial defining the system. A consequence of the distribution property is that it reduces the study of $H(2)$ to the study of the maximum number of limit cycles surrounding one singularity.

Keywords: quadratic system, limit cycle, 16th Hilbert problem

MSC 2020: 34C05, 34C07

1 Introduction

This article concerns the study of limit cycles in so-called *quadratic systems*

$$\begin{aligned}\frac{dx(t)}{dt} &= P_2(x(t), y(t)) = a_{00} + a_{10}x(t) + a_{01}y(t) + a_{20}x^2(t) + a_{11}x(t)y(t) + a_{02}y^2(t), \\ \frac{dy(t)}{dt} &= Q_2(x(t), y(t)) = b_{00} + b_{10}x(t) + b_{01}y(t) + b_{20}x^2(t) + b_{11}x(t)y(t) + b_{02}y^2(t),\end{aligned}\tag{1.1}$$

where $(x, y) \in \mathbb{R}^2$, $t \in \mathbb{R}$, $a_{ij}, b_{ij} \in \mathbb{R}$.

For notational convenience, we will drop the dependence on the time parameter t in the variables x and y .

In the past century, many results have been obtained for quadratic systems, in particular in China and the former Soviet Union. Several excellent review papers exist, see the study by Artés, et al. [1] for the most recent updates and studies by Coppel [7] and Ye et al. [14]. Finding the maximum number of limit cycles in these systems is one of the outstanding open problems in differential equations, being part of Hilbert's sixteenth problem [1], asking for the (best) upper bound on the number of limit cycles for polynomial systems of degree n denoted by $H(n)$, where n is the maximum degree of the polynomials. Here, we consider the case $H(2)$. Currently, it is known that $H(2) \geq 4$, because examples exist of quadratic systems with at least four limit cycles [2,11].

For a detailed description of the rich history of quadratic systems [14]. In this article, the basic properties that will be used are the following:

- [13,16] Limit cycles in a quadratic system surround exactly one singularity which necessarily is a focus.
- [7,8] A quadratic system can have at most two foci.
- [7,8] A quadratic system can only have at most two nests of limit cycles, where a *nest* denotes the collection of nested limit cycles surrounding one singularity.

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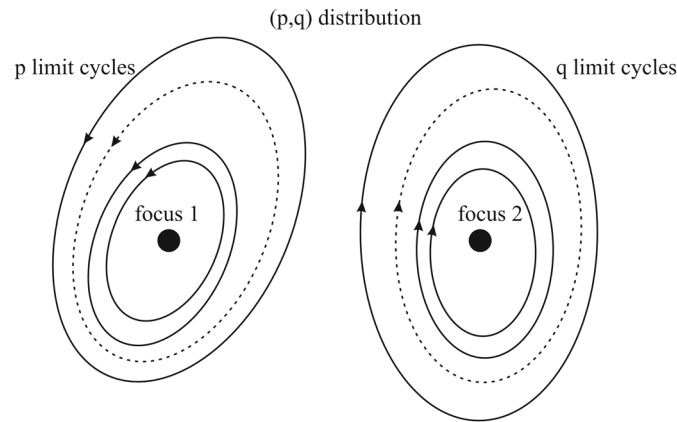


Figure 1: Two nests of limit cycles in a quadratic system surrounding two foci.

Define the number of limit cycles in a quadratic system with two nests as (p, q) , where p (q) is the number of limit cycles surrounding the first (second) focus. See Figure 1 for a typical phase portrait of two nests in a quadratic system.

The first famous examples of a nest distribution of limit cycles were given in the 70's in China where $p \geq 3$, $q \geq 1$ [2,11]. Several attempts were made in the following years to prove that a $(2, 2)$ distribution is not possible [15] or to find a $(3, 2)$ or even a $(3, 3)$ distribution. These attempts remained futile.

Finally, Zhang proved in a series of remarkable papers published in China that the only possible distributions for a quadratic system are $(p_1, 0)$ or $(p_2, 1)$, implying that $H(2) = \max\{p_1 + 1, p_2\}$. His collection of papers was summarized in an English publication [22]. The aim of this article is to clarify and simplify his proof, but moreover to fill in some essential gaps in the study by Zhang [22]. Although we believe that the idea of the proof is essentially correct, some details were omitted which apparently were not in the Chinese papers either.

It was mentioned in Section 2.2 of the book [1] that “this result of Zhang Pingguang has so far not been checked by the mathematical community.” Our aim is to show that the conclusion of Zhang [22] for the distribution of limit cycles is essentially correct.

For readability purposes, we will define the *distribution property* as follows:

Definition 1.1. By the expression *distribution property* we mean: two nests of limit cycles where in (at least) one nest at most one limit cycle exists, i.e., $(n, 0)$ or $(n, 1)$, with $n \in \{0\} \cup \mathbb{N}$.

One of the critical elements in the proof of the distribution property for quadratic systems is to check conditions in a uniqueness theorem for limit cycles in a Liénard system. On page 456 in the study by Zhang [22], the check of the monotonicity of a critical function $\frac{f(x)F(x)}{g(x)}$ was omitted with the justification that it was easy to see. It was not mentioned in the original Chinese papers either, and in our opinion, the suggested way of proving the monotonicity is not the correct way. In this article, we will show the details of this critical part of the proof.

Some of the theorems that were used by Zhang [22] appeared only in the Chinese papers and are not readily accessible for other audiences. Moreover, some cases, like the case of quadratic systems with four real singularities, can be proved in a much simpler way as this article intends to show.

The structure of this article is as follows.

First, we identify in Section 2 which quadratic systems can contain nests of limit cycles. For this purpose, we introduce a new canonical form which makes it easy to find finite and infinite singularities in quadratic systems and to find parameters corresponding to rotated vector fields in relevant parts of the phase plane.

In Section 3, the relevant theorems for Liénard systems are listed. There are essentially two theorems needed for the proof of the distribution property: one for proving the non-existence of limit cycles and one for proving the uniqueness of limit cycles. The two theorems are classical results in the field of Liénard systems.

The non-existence theorem is one of the strongest theorems available. It was used by Li [10] in the proof of non-existence of limit cycles in a quadratic system with a third-order weak focus. The main condition that needs to be verified is the non-existence of solutions to a system of two algebraic equations with two variables.

The uniqueness theorem is a classical result often referred to as Coppel's theorem, for which two conditions need to be checked. Firstly, the same system of algebraic equations as in the non-existence theorem needs to have at most one solution. Secondly, on a restricted interval, the derivative of a certain function $\frac{f(x)F(x)}{g(x)}$ needs to have fixed sign.

Since in practice it is impossible to solve the system of algebraic equations, it is not easy to prove an upper bound of 0 or 1 for the number of solutions. For these purposes, we reformulate the two theorems in terms for which the conditions are easier to check, i.e., they reduce to the study of a function of one variable only.

In three sections, we will prove the distribution property for different configurations of the singularities.

In Section 4, we prove the distribution property for quadratic systems with two finite real strong foci, one pair of complex singularities and (at least) one elementary saddle at infinity, and no invariant line. This case is the most difficult one and contains the examples of four limit cycles in a (3, 1)-distribution.

In Section 5, we do the same for quadratic systems with four real singularities, simplifying the proofs of Zegeling and Kooij [17] and Zhang [22].

In Section 6, using the theory of rotated vector fields, we discuss the distribution property for the remaining non-generic cases: quadratic systems with exactly three finite real singularities, exactly two finite real singularities, with a higher order finite real singularity, with a weak focus or with a higher order saddle at infinity, i.e., the cases not covered by Sections 4 and 5.

Finally, in Section 7, we collect the results of the previous sections and prove the main result of this article that any quadratic system has the distribution property of Definition 1.1.

Theorem 1.2. *A quadratic system (1.1) can only have an $(n, 1)$ or $(n, 0)$ distribution of limit cycles where $n \in \{0\} \cup \mathbb{N}$.*

2 Structure of nests of limit cycles in quadratic systems

To study the two possible nests in a quadratic system, we categorize the possible configurations of the singularities.

One of the classical results for quadratic systems is the following result by Sakhova [12].

Lemma 2.1. *If a quadratic system (1.1) has a real finite singularity, then there is a straight line forming an isocline.*

This lemma serves as the basis of the Chinese classification Class I, II, and III of quadratic systems [14]:

$$\begin{aligned}\frac{dx}{dt} &= \delta x - y + lx^2 + mxy + ny^2, \\ \frac{dy}{dt} &= x(1 + ax + by),\end{aligned}\tag{2.1}$$

where $\delta, l, m, n, a, b \in \mathbb{R}$. For $a = b = 0$, this becomes class I, where the line $x = 0$ forms the 0-isocline. For $b = 0, a \neq 0$, this becomes class II, where the pair of parallel lines $x = 0, x = -\frac{1}{a}$ form the 0-isocline. For $b \neq 0, a \neq 0$, this becomes class III, where the pair of intersecting lines $x = 0, 1 + ax + by = 0$ form the 0-isocline.

We extend Sakhova's result to the case of two foci.

Lemma 2.2. *If a quadratic system (1.1) has two real foci, then the collection of isoclines contains a pair of intersecting straight lines.*

Proof. Intuitively the statement of the lemma is not surprising, because it is well known that the line connecting two singularities in (1.1) [14] is an isocline and to have two singularities with the same index (in this case two anti-saddles) on the connecting line, we would expect another branch of the isocline to intersect it.

After an affine transformation we place the two foci at $O(0, 0)$ and $N(0, 1)$. The quadratic system takes the form.

$$\begin{aligned}\frac{dx}{dt} &= P_2(x, y) = a_{10}x - a_{02}y + a_{20}x^2 + a_{11}x(t)y + a_{02}y^2, \\ \frac{dy}{dt} &= Q_2(x, y) = b_{10}x - b_{02}y + b_{20}x^2 + b_{11}x(t)y + b_{02}y^2.\end{aligned}\quad (2.2)$$

Here, $a_{02} \neq 0$, because if it were zero, then the line $x = 0$ would be a solution and the two singularities O and N would not be foci. The line $x = 0$ is an isocline for the slope $\frac{dy}{dx} = \frac{b_{02}}{a_{02}}$. All other branches of the isocline with this slope satisfy:

$$a_{02}b_{10} - b_{02}a_{10} + x(a_{02}b_{20} - b_{02}a_{20}) + y(a_{02}b_{11} - b_{02}a_{11}) = 0. \quad (2.3)$$

The product of the eigenvalues of the linearized system at O is:

$$P_0 = a_{02}b_{10} - a_{10}b_{02},$$

which is positive according to the assumption that O is a focus. The product of the eigenvalues of the linearized system at N is given as follows:

$$P_1 = -(a_{02}b_{11} - b_{02}a_{11}) - (a_{02}b_{10} - a_{10}b_{02}) \equiv -S - P_0.$$

For this product to be positive as well, necessarily $S < 0$, i.e., the line defined through (2.3) is a line intersecting the line $x = 0$, proving the lemma.

We note that the statement of the lemma could be made stronger by writing “two anti-saddles” or even more general “two singularities with the same index”, because the proof only used that the product of the eigenvalues at the two singularities has the same sign. $\square$

One consequence of the lemma is that any quadratic system with two foci can be brought into the form of class III of the Chinese classification. This fact was implicitly used by Zhang [22].

An additional property is that the line defined by (2.3) intersects $x = 0$ between the two foci, i.e., for $0 < \bar{y} < 1$. This follows from: $\bar{y} = -\frac{P_0}{S} > 0$ and $1 - \bar{y} = -\frac{P_1}{S} > 0$.

Next we use Lemma 2.2 to bring the quadratic system with two foci into a useful canonical form, introduced in the study by Zegeling [18]. An affine transformation applied to (2.2) brings the isocline $\frac{dy}{dx} = \frac{b_{02}}{a_{02}}$ onto the coordinate axes $x = 0$ and $y = 0$, i.e., $x = 0$ is moved to $y = 0$ and the line defined through (2.3) is brought onto $x = 0$. Here, we will impose that the line defined through (2.3) is not a solution to the system, because then the quadratic system (1.1) is well known to have at most one limit cycle and a distribution of limit cycles cannot occur (see Section 4, Lemma 4.2).

We obtain the following quadratic system:

$$\begin{aligned}\frac{dx(t)}{dt} &= a_{11}xy + f(x, y), \\ \frac{dy(t)}{dt} &= b_{11}xy + f(x, y),\end{aligned}\quad (2.4)$$

where $f(x, y) = (1 - \frac{x}{x_1})(1 - \frac{x}{x_2}) + p_1y + p_2y^2$, $(a_{11}, b_{11}, x_1, x_2, p_1, p_2) \in \mathbb{R}^6$. Here, $x_1 < 0$ and $x_2 > 0$ correspond to the x -coordinates of the two foci positioned on the x -axis. They are anti-saddles if and only if $a_{11} > b_{11}$, because the product of the eigenvalues for the singularity at $x = x_1 < 0$ ($x = x_2 > 0$) at the linearized system is $(a_{11} - b_{11})\frac{x_2 - x_1}{x_2} ((a_{11} - b_{11})\frac{x_1 - x_2}{x_1})$. In this canonical form, all finite singularities reside on the coordinate axes. In addition to the two foci, other finite singularities might exist on the y -axis satisfying:

$$1 + p_1y + p_2y^2 = 0. \quad (2.5)$$

Using this equation, we can distinguish the following cases:

Lemma 2.3. *A quadratic system (1.1) with two nests of limit cycles can be characterized by the following mutually exclusive subsets:*

- (1) (generic) *There are exactly two foci and one pair of finite complex singularities.*
- (2) (generic) *There are exactly four real finite singularities of which two are foci.*
- (3) (non-generic)
 - *There are exactly three real finite singularities of which two are foci and the third is a singularity with eigenvalue equal to zero.*
 - *There are exactly three real finite singularities of which two are foci and the third is an elementary saddle.*
 - *There are exactly two foci and no other finite singularities.*

Proof. The lemma follows immediately from system (2.4) which contains all quadratic systems with two foci according to Lemma 2.2 and the fact that a quadratic system with an invariant line cannot have a distribution of limit cycles. The cases of 2.3 correspond to the different sets of singularities on the y -axis satisfying equation (2.5):

- (1) $p_1^2 - 4p_2 < 0$.
- (2) $p_1^2 - 4p_2 > 0$.
- (3)
 - $p_1 \neq 0, p_2 = 0$.
 - $p_1^2 - 4p_2 = 0$.
 - $p_1 = 0, p_2 = 0$.

□

To interpret the cases in Lemma 2.3, we observe that only two cases are generic: cases 1 and 2. The other three cases collected under Case 3 are labelled as non-generic and can be perturbed by changing the parameter p_2 , which will move them into one of the cases 1 and 2. Therefore, our strategy for proving the distribution property for quadratic systems is to give a detailed proof for the two generic cases and then use a bifurcation argument to prove the other cases.

3 Liénard theorems

The proofs in this article rely heavily on the application of theorems in so-called Liénard systems. The standard Liénard form is:

$$\begin{aligned}\frac{dx}{dt} &= y, \\ \frac{dy}{dt} &= -g(x) - f(x)y.\end{aligned}\tag{3.1}$$

It is well known that the Liénard form (3.1) is equivalent to the generalized form [14,19]:

$$\begin{aligned}\frac{dx}{dt} &= F(x) - y, \\ \frac{dy}{dt} &= g(x),\end{aligned}\tag{3.2}$$

where $F(x) = \int f(\bar{x})d\bar{x}$.

For systems (3.1) and (3.2), the following theorems are essential for the proof of the distribution property.

Theorem 3.1. (Non-existence) [19] *Suppose system (3.1) with differentiable $F(x) = F_{x_g}(x) = \int_{x_g}^x f(\bar{x})d\bar{x}$, and continuous $g(x), f(x)$, satisfies the following conditions on the interval $r_1 < x < r_2$.*

- (i) $x_g \in (r_1, r_2)$ exists such that $(x - x_g)g(x) > 0$, for $x \neq x_g$, and $g(x_g) = 0$,

- (ii) $x_f \in (r_1, r_2)$ exists such that $(x - x_f)f(x) < 0$ for $x \neq x_f$, and $f(x_f) = 0$, $x_f \neq x_g$,
 (iii) the system of equations

$$\frac{f}{g}(x_1) = \frac{f}{g}(x_2), \quad F_{x_g}(x_1) = F_{x_g}(x_2)$$

has no solution $(x_1 = \bar{x}_1, x_2 = \bar{x}_2)$ satisfying: $r_1 < \bar{x}_1 < x_g < \bar{x}_2 < r_2$.

Then (3.1) does not have limit cycles in the strip $r_1 < x < r_2$.

Remark 3.2. The first two conditions (i) and (ii) have the following interpretation:

- There is a unique anti-saddle.
- The divergence has a unique zero where it changes sign.
- $x_f \neq x_g$, i.e., the system does not have a weak focus or centre.

In the Liénard systems considered in this article these conditions can easily be verified to be true in each region where a nest of limit cycles can occur.

For proving the uniqueness of limit cycles in a Liénard system, the following theorem will be used, which is sometimes referred to as Coppel's theorem even though it already occurred in the Chinese literature before the publication of his article.

Theorem 3.3. (Uniqueness) [6,9,19,21,22] Suppose system (3.1) with differentiable $F(x) = F_{x_g}(x) = \int_{x_g}^x f(\bar{x})d\bar{x}$, and continuous $g(x)$, $f(x)$, satisfies the following conditions on the interval $r_1 < x < r_2$.

- (i) $x_g \in (r_1, r_2)$ exists such that $(x - x_g)g(x) > 0$, for $x \neq x_g$, and $g(x_g) = 0$,
 (ii) $x_f \in (r_1, r_2)$ exists such that $(x - x_f)f(x) < 0$ for $x \neq x_f$, and $f(x_f) = 0$, $x_f \neq x_g$,
 (iii) the system of equations

$$\frac{f}{g}(x_1) = \frac{f}{g}(x_2), \quad F_{x_g}(x_1) = F_{x_g}(x_2), \quad (3.3)$$

has exactly one solution $(x_1 = \bar{x}_1, x_2 = \bar{x}_2)$ satisfying: $r_1 < \bar{x}_1 < x_g < \bar{x}_2 < r_2$,

- (iv) either (a) $(\frac{fF_{x_g}}{g})'(x) < 0$ for $r_1 < x < \bar{x}_1$ or (b) $(\frac{fF_{x_g}}{g})'(x) > 0$ for $\bar{x}_2 < x < r_2$, where $(\bar{x}_1, \bar{x}_2)$ is the solution to (3.3).

In addition, in case (a) (case (b)), the condition $F(r_2) \leq F(r_1)$ ($F(r_1) \leq F(r_2)$) needs to hold if $x_g < x_f$ ($x_g > x_f$).

Then (3.1) has at most one limit cycle in the strip $r_1 < x < r_2$, and if it exists, it is hyperbolic.

See Figure 2 for the situation of condition (iv). The sign of $f(x)$ in the second condition is not critical. The condition can be replaced by $(x - x_f)f(x) < 0$, but in this article, we will only encounter the case of the theorem, so we omitted it.

Remark 3.4. The rather ominous looking condition on $F(r_1)$ and $F(r_2)$ in the last condition is due to a technical complication in the proof of the theorem [9]. In practice, it does not seem to cause major issues. In this article,

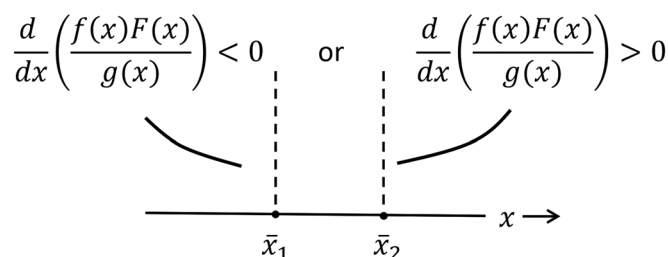


Figure 2: Condition (4) of Theorem 3.3.

at the end points of the interval, r_1 and r_2 $F(x)$ will tend to minus infinity and the condition is satisfied because of the equality sign in $F(r_2) \leq F(r_1)$ and $F(r_1) \leq F(r_2)$.

In condition (3), if there is no solution, then Theorem 3.1 shows that there are no limit cycles.

For practical purposes, the conditions can be rewritten. Essentially we rewrite the study of the condition involving the algebraic equation with two variables (3.3), in terms of the study of a function $Z(x)$ of one variable x only.

Theorem 3.5. (Non-existence adjusted) Suppose system (3.1) with differentiable $F(x) = F_{x_g}(x) = \int_{x_g}^x f(\bar{x})d\bar{x}$, and continuous $g(x)$, $f(x)$, satisfies the following conditions on the interval $r_1 < x < r_2$, where we define

$$Z(x) = \frac{f(x)(F_{x_g}(x) + K)}{g(x)},$$

(i) $(x - x_g)g(x) > 0$ for $x \neq x_g$,

(ii) $(x - x_f)f(x) < 0$, $x \neq x_f$, $x_f \neq x_g$,

(iii) $\exists c, K \in \mathbb{R}$ such that $Z(x) = c$ does not have a solution and $F_{x_g}(x_g) + K \neq 0$.

Then (3.1) does not have limit cycles in the strip $r_1 < x < r_2$.

Proof. According to condition (i) $g(x)$ has exactly one zero x_g , and according to $F_{x_g}(x_g) + K \neq 0$ from condition (iii), the function $Z(x)$ has exactly one vertical asymptote at $x = x_g$, where the function tends to $-\infty$ on one side and to $+\infty$ on the other side. It follows that for $0 < \varepsilon \ll 1$, $(Z(x_g + \varepsilon) - c)(Z(x_g - \varepsilon) - c) < 0$, where c is the constant in condition (iii). Since $Z(x)$ is continuous for $x \neq x_g$, it follows that $(Z(x_1) - c)(Z(x_2) - c) < 0$ for all $r_1 < x_1 < x_g < x_2 < r_2$, because according to condition (iii) $Z(x) \neq c$. Therefore, the equation $Z(x_1) = Z(x_2)$ with $r_1 < x_1 < x_g < x_2 < r_2$ does not have a solution.

This implies that the set of equation (3.3) has no solutions $r_1 < x_1 < x_g < x_2 < r_2$ either. First, addition of K to the first equation on both sides gives $F(x_1) + K = F(x_2) + K$. Then multiplying this equation with the second equation leads to $Z(x_1) = Z(x_2)$, for which we have just proved that no solutions exist.

Therefore, the third condition (iii) of Theorem 3.1 holds, proving that no limit cycles exist. $\square$

In practice, the interpretation of this non-existence theorem is that we will look for a horizontal gap in the graph of $y = Z(x)$.

Theorem 3.6. (Uniqueness adjusted) Suppose system (3.1) with differentiable functions $F(x) = F_{x_g}(x) = \int_{x_g}^x f(\bar{x})d\bar{x}$, and continuous $g(x)$, $f(x)$ satisfies the following conditions on the interval $r_1 < x < r_2$ where we define

$$Z(x) = \frac{f(x)(F_{x_g}(x) + K)}{g(x)},$$

(i) $x_g \in (r_1, r_2)$ exists such that $(x - x_g)g(x) > 0$, for $x \neq x_g$, and $g(x_g) = 0$;

(ii) $x_f \in (r_1, r_2)$ exists such that $(x - x_f)f(x) < 0$ for $x \neq x_f$, and $f(x_f) = 0$, $x_f \neq x_g$;

(iii) $\exists K \in \mathbb{R}$ such that $Z'(x)$ has the same (positive or negative) sign for every $x \in (a_1, b_1) \cup (a_2, b_2)$. This is the collection of intervals $a_1 < x_1 < b_1$, $a_2 < x_2 < b_2$ for which $Z(x_1) = Z(x_2)$ has a solution pair (x_1, x_2) , where $r_1 \leq a_1 < b_1 < x_g < a_2 < b_2 \leq r_2$,

(iv) either (a) $(\frac{fF_{x_g}}{g})'(x) < 0$ for $r_1 < x < \bar{x}_1$ or (b) $(\frac{fF_{x_g}}{g})'(x) > 0$ for $\bar{x}_2 < x < r_2$, where $(\bar{x}_1, \bar{x}_2)$ is the solution to $F_{x_g}(x_1) = F_{x_g}(x_2)$, $Z(x_1) = Z(x_2)$. In addition, in case (a) (case (b)), the condition $F(r_2) \leq F(r_1)$ ($F(r_1) \leq F(r_2)$) needs to hold if $x_g < x_f$ ($x_g > x_f$).

Then (3.1) has at most one limit cycle in the strip $r_1 < x < r_2$, and if it exists, it is hyperbolic.

Proof. Since all conditions are the same as in Theorem 3.3, except for condition (iii), we need to show that condition (iii) of Theorem 3.3 holds true.

The equation $Z(x_1) = Z(x_2)$ defines a curve γ_Z in the (x_1, x_2) plane with one component, because the function $Z(x)$ is continuous for $x \neq x_g$. At any point (x_1, x_2) of the curve, the slope of the tangent line $\frac{dx_2}{dx_1}$ is

given by $\frac{Z'(x_1)}{Z'(x_2)}$. Since $Z'(x)$ has the same sign for all x defining γ_Z according to condition (iii) in Theorem 3.6, the slope of the tangent line is always positive. Therefore, the curve γ_Z is monotonically decreasing as a function of x_1 .

Suppose $x_f > x_g$ (the case $x_g > x_f$ can be handled in exactly the same way), then the function $\frac{f(x)}{g(x)}$ defining the second equation in the set of equation (3.3) has the same sign on both sides of $x = x_g$ only for $x < x_g$ and $x > x_f$, i.e., $x_1 < x_g$ and $x_g < x_f < x_2$. It follows that $F(x)$ in $F(x_1) = F(x_2)$ satisfies $F'(x_1) > 0$ and $F'(x_2) < 0$, i.e., the tangent line at any point on the curve γ_F satisfies $\frac{dx_2}{dx_1} = \frac{F'(x_1)}{F'(x_2)} = \frac{f(x_1)}{f(x_2)} < 0$. The continuous curve γ_F is monotonically decreasing as a function of x_1 .

Next we rewrite the set of equation (3.3) in condition (iii) of Theorem 3.3 (see the proof of Theorem 3.5) as $F_{x_g}(x_1) = F_{x_g}(x_2)$, $Z(x_1) = Z(x_2)$ by adding K to the equation $F(x_1) = F(x_2)$ on both sides and then multiplying it with the second equation. These two equations define two curves γ_F and γ_Z in the (x_1, x_2) -plane for which we just proved that they are monotonically decreasing and increasing, respectively. Both have only one component and accordingly we conclude that they have at most one intersection, corresponding to a solution of the set of equations in condition (3.3).

Note that the uniqueness of the solution $(\bar{x}_1, \bar{x}_2)$ is used in condition (iv). If the solution does not exist, then according to Theorem 3.1, no limit cycles exist. $\square$

Remark 3.7. Theorem 3.6 looks more complicated than Theorem 3.3. However, in practice, the constant K can be chosen in such a way that the new function $Z(x) = \frac{f(x)(F_{x_g}(x) + K)}{g(x)}$ has a nice structure. Conditions (iii) and (iv) are then relatively easy to check compared to a direct study of the system of equation (3.3) in Theorem 3.3. The investigation of a system of two nonlinear equations with two variables reduces to the study of a function of one variable.

In our case, we will see that the function $Z(x)$ can be chosen to be a rational function of order 4. The application of Theorems 3.5 and 3.6 then essentially boils down to detecting gaps in the graph of $y = Z(x)$ or showing monotonicity of the rational function $Z(x)$.

4 Quadratic systems with two strong foci and two complex singularities

In this section, we will study the first of the two generic cases of quadratic system with two foci: Case (1) in Lemma 2.3. This is the case where the first two examples of (3, 1) distributions in quadratic systems were found [2,11]. It is the most difficult and most critical case for the quadratic case in Hilbert's sixteenth problem.

Since the proof of the theorems depends on the use of theorems in the form of a Liénard equation, the natural starting point of the proof is to study quadratic systems in a form which allows an easy transformation into Liénard form.

Zhang [22] used class III (2.1) in the Chinese classification of quadratic systems as a starting point, which has one anti-saddle positioned at the origin of the phase plane. For the study of the distribution of limit cycles, this is not necessary. In fact, it is more natural to keep the coordinates of the two foci depending on parameters.

The natural way to enforce this is to use the classification of quadratic systems with limit cycles by Cherkas [3]. Moreover, in his classification, the transformation to a Liénard system is easier than in class III, and the two nests appear in two half planes separated by the y -axis. The canonical form (2.4) provides a good platform to study finite and infinite singularities in quadratic systems, but its transformation into a Liénard system is rather cumbersome and is not easy to use in the application of Liénard theorems.

4.1 Canonical form for the quadratic system

Lemma 4.1. [3] *A quadratic system (1.1) with limit cycles can be transformed through affine transformations into one of the two following forms:*

Class I

$$\begin{aligned}\frac{dx}{dt} &= a_{00} + xy, \\ \frac{dy}{dt} &= Q_2(x) + Q_1(x)y + Q_0y^2,\end{aligned}\tag{4.1}$$

Class II

$$\begin{aligned}\frac{dx}{dt} &= y + x^2, \\ \frac{dy}{dt} &= Q_2(x) + Q_1(x)y + Q_0y^2,\end{aligned}\tag{4.2}$$

where

$$\begin{aligned}Q_2(x) &= a_0 + a_1x + a_2x^2, \\ Q_1(x) &= b_0 + b_1x, \\ Q_0 &\in \mathbb{R}.\end{aligned}\tag{4.3}$$

Proof. The derivation of these classes is straightforward. Choose any singularity at infinity and apply a linear transformation to bring it to the end of $x = 0$. This means that in the original quadratic system (1.1) the coefficient a_{02} is 0. There are two cases to consider: $a_{11} \neq 0$ and $a_{11} = 0$, leading to class I and class II of the Cherkas classification. Suppose $a_{11} \neq 0$. Apply a transformation $a_{20}x + a_{11}y = u$ to (1.1) with $a_{02} = 0$. This brings the $\frac{dx}{dt}$ equation into a form $a_{00} + \tilde{a}_{10}x + \tilde{a}_{01}u + xu$. Next we eliminate the parameter $\tilde{a}_{01}$ by setting $\bar{x} = x + \tilde{a}_{01}$, leading to $\frac{d\bar{x}(t)}{dt} = \tilde{a}_{00} + \tilde{a}_{10}\bar{x} + \bar{x}u$. Finally, we can eliminate $\tilde{a}_{10}$ by shifting the u -variable: $\bar{u} = u + \tilde{a}_{10}$, which leads to:

$$\frac{d\bar{x}(t)}{dt} = \tilde{a}_{00} + \bar{x}\bar{u}.$$

For the case $a_{11} = 0$, a similar procedure can be applied giving (4.2). □

For our purposes, we will assume that $\tilde{a}_{00} \neq 0$ because if the parameter were 0, then the quadratic system would have an invariant line. This is a system known to have at most one limit cycle.

Lemma 4.2. [5,6] *A quadratic system (1.1) with an invariant line has at most one limit cycle. In particular, any non-trivial distribution of limit cycles takes the form (1, 0).*

Therefore, after a scaling in the variables, we can take $a_{00} = 1$ in system (4.1).

To study the limit cycle distributions, we first need to decide in which of the two systems (4.1) and (4.2) in Lemma 4.1 two foci and a pair of complex singularities can occur.

Lemma 4.3. *Only system (4.1) of Lemma 4.1 contains the case of two foci and a pair of complex singularities. This case cannot occur in system (4.2).*

Proof. The intuitive reason why class II (4.2) cannot have just two finite foci while class I (4.1) can is explained below.

In (4.2), all singularities lie on the parabola $y = -x^2$ and lie on the same branch of the curve given by $\frac{dx}{dt} = 0$, while for (4.1), the curve given by $\frac{dx}{dt} = 0$ is a hyperbola given by two branches, one for $x < 0$ and one for $x > 0$.

Neighbouring singularities on the same branch of the curve $\frac{dx}{dt} = 0$ typically will have opposite index. This was the same idea behind the proof of Lemma 2.2.

Therefore, it is impossible to have exactly two finite foci on one branch, because their existence would imply a third singularity (i.e., a saddle) lying in-between them. Since (4.2) has exactly one branch for $\frac{dx}{dt} = 0$, it is impossible to obtain exactly two finite foci.

This obstacle does not occur in class I (4.1): the singularities lie on two non-connected branches of a hyperbola and the same argument does not apply, because two singularities on different branches are separated by the isocline (and line without contact) $x = 0$.

Mathematically this can be seen through the following calculation for (4.2). A straightforward computation shows that the singularities of the system satisfy:

$$H(x) \equiv Q_2(x) - Q_1(x)x^2 + Q_0x^4 = 0.$$

The product of eigenvalues at the singularities is given by:

$$2x(Q_1(x) - 2Q_0x^2) - Q_2'(x) + x^2Q_1'(x) = -H'(x).$$

It follows that the sign of the product of eigenvalues of adjacent singularities is alternating as expected. At two points x_1, x_2 satisfying $H(x_i) = 0$ and $H(x) \neq 0$ for $x \in (x_1, x_2)$ necessarily we have $H'(x_1)H'(x_2) < 0$ for elementary singularities. This shows that it is impossible to have two finite foci without a saddle positioned in-between them. $\square$

Remark 4.4. Obviously this lemma implies that we do not have to consider class II for the study of limit cycle distributions in quadratic systems with exactly two finite foci and a pair of complex finite singularities.

The condition we need to impose is that a quadratic system has exactly two (real) finite foci and two complex singularities. One observation is that if the system has two finite anti-saddles, then an index argument implies that at infinity at least one singularity needs to exist of saddle type:

Lemma 4.5. *A quadratic system with two finite foci has a (at least one) singularity at infinity, which is of saddle-type. Generically, it is an elementary saddle. In the non-generic case, it is a higher order saddle with one zero eigenvalue.*

Proof. Consider system (2.4), Case (1) of Lemma 2.3. After a rescaling of variables, we may assume that $p_2 = 1$. To study the singularities at infinity, we apply the Poincaré transformation $x = \frac{1}{z}$, $y = \frac{u}{z}$ and bring the system (after a scaling of time) into the form:

$$\begin{aligned} \frac{du}{dt} &= z(1-u)(z + p_1u - \lambda(x_1 + x_2)) - h_1(u), \\ \frac{dz}{dt} &= z(h_2(u) + z(\lambda(x_1 + x_2) - p_1u) - z^2), \end{aligned} \quad (4.4)$$

where

$$\begin{aligned} h_1(u) &= u^3 + u^2(a_{11} - 1) + u(\lambda - b_{11}) - \lambda, \\ h_2(u) &= (-u^2 + ua_{11} + \lambda), \\ \lambda &= \frac{1}{x_1x_2}. \end{aligned} \quad (4.5)$$

Singularities at infinity satisfy $z = 0$ and $h_1(u) = 0$. For any such singularity, the function $h_2(u)$ can be rewritten as $\frac{(a_{11} - b_{11})u}{u - 1}$, which shows that $h_2(u) \neq 0$ when $h_1(u) = 0$, because $u = 0$ is not a singularity at infinity.

The equation $h_1(u)$ has a negative solution u^* , because the constant in the cubic equation satisfies $\lambda < 0$. Either $h_1'(u) < 0$ or $h_1'(u) = h_1''(u) = 0$, $h_1'''(u) > 0$. In the first generic case $h_1'(u) < 0$, the singularity is an elementary saddle, because the product of eigenvalues is given by $h_1'(u)h_2(u) < 0$. Here, we also used that $u^* < 1$, because $h_1(1) > 0$.

In the second case $h_1'(u) = h_1''(u) = 0$, $h_1'''(u) > 0$, there is exactly one singularity at infinity and since we assumed that there are exactly two finite foci, an index argument shows that this singularity is a higher order saddle, i.e., with zero eigenvalues. For our proofs, we will not need to know the exact character of this singularities, because it corresponds to a non-generic case where we will apply a bifurcation argument to prove the distribution property. $\square$

An important property of the system is that a line without contact exists which limit cycles cannot cross.

Lemma 4.6. *In system (4.1), limit cycles cannot cross the line $x = 0$, i.e., limit cycles lie either completely in $x < 0$ or in $x > 0$. A distribution of limit cycles (n, m) consists of n (m) limit cycles lying entirely in $x < 0$ ($x > 0$).*

Proof. This follows from the observation that $\frac{dx}{dt} = 1 > 0$ for $x = 0$. $\square$

In the proof of the distribution property of limit cycles for class (4.1), we will not need to impose necessary and sufficient conditions on the parameters. Some necessary conditions though will drastically reduce the cases to be considered in the following.

Lemma 4.7. *For system (4.1) with exactly two finite foci and two complex finite singularities and an elementary saddle at infinity, and no invariant line, the following parameter restrictions can be made: $a_{00} = 1$, $a_2 = -1$, $0 < Q_0 < 1$.*

Proof. System (4.1) has seven parameters. Take $x \rightarrow \mu x$, $y \rightarrow \frac{1}{\mu}y$, $t \rightarrow \frac{1}{\mu}t$ and choose μ to set the parameter a_2 equal -1 .

The x -coordinates of the finite singularities satisfy the following equation:

$$H(x) \equiv Q_2(x) - \frac{Q_1(x)}{x} + \frac{Q_0}{x^2} = 0$$

with $H'(x) = Q_2'(x) - \frac{Q_1'(x)}{x} + \frac{Q_1(x)}{x^2} - 2\frac{Q_0}{x^3}$. The product of eigenvalues of finite singularities is easily shown to be equal to:

$$-\frac{H'(x)}{x}.$$

Similar to the argument in the proof of Lemma 4.3 we conclude that adjacent singularities for $(x < 0)$ or $(x > 0)$ have an opposite index. Therefore, to have two finite foci, one focus needs to lie in the half-plane $x < 0$ and one focus in $x > 0$. Both half-planes are separated by the line without contact $x = 0$ as indicated by Lemma 4.6, i.e., limit cycles cannot cross this line. We label the x -coordinates of the singularities by $x_g^- < 0$ and $x_g^+ > 0$, respectively.

Next we impose that the singularities are anti-saddles. This means that $H'(x_g^+) < 0$ and $H'(x_g^-) > 0$. Since we do not allow other singularities to exist, we obtain as a necessary condition that the parameter a_2 in $P_2(x)$ needs to be negative: this ensures that $\lim_{|x| \rightarrow \infty} H(x) = -\infty$ giving the correct sign of the derivative $H'(x)$ at the two zeroes x_g^- and x_g^+ of $H(x)$.

A necessary condition on the parameters can be imposed by requiring that the behaviour of $H(x)$ near $x = 0$ does not contradict the requirement that $H'(x_g^+) < 0$ and $H'(x_g^-) > 0$. Explicitly we need that $\lim_{|x| \rightarrow 0} H(x) = \infty$. This is achieved by requiring that the parameter Q_0 is positive.

Next we impose that the point at infinity at the end of $x = 0$ is an elementary saddle. After the standard transformation to the Poincaré sphere $x = \frac{1}{v}$, $y = \frac{u}{v}$, the linearization of the system near the singularity $(u = 0, v = 0)$ shows that it is an elementary saddle if and only if $Q_0 < 1$.

Since according to Lemma 4.2 a quadratic system with an invariant line (i.e., $a_{00} = 0$ in (4.1)) can only have a $(1, 0)$ distribution, we remove this case from our investigation by setting $a_{00} = 1$ after some affine transformations. $\square$

4.2 Transformation to Liénard form

Next we transform system (4.1) into the Liénard form (3.1).

Lemma 4.8. *System (4.1) under the restrictions of Lemma 4.7 can be transformed into the standard Liénard form (3.1) with*

$$\begin{aligned} g(x) &= W_4(x) \frac{|x|^{-2Q_0-2}}{x}, \\ f(x) &= W_2(x) \frac{|x|^{-Q_0-1}}{x}, \end{aligned} \quad (4.6)$$

where

$$\begin{aligned} W_4(x) &\equiv -x^2(a_0 + a_1x - x^2) + x(b_0 + b_1x) - Q_0, \\ W_2(x) &\equiv -b_1x^2 - b_0x + 1 + 2Q_0. \end{aligned}$$

The function $F(x) = \int_{x_g}^x f(\bar{x}) d\bar{x}$ is given by

$$\begin{aligned} \text{for } x > 0 : F_+(x) &= \int_{x_g^+}^x f(\bar{x}) d\bar{x} = c_+ + F_0(x), \\ \text{for } x < 0 : F_-(x) &= \int_{x_g^-}^x f(\bar{x}) d\bar{x} = c_- + F_0(x), \end{aligned} \quad (4.7)$$

where

$$F_0(x) = |x|^{-Q_0-1} \left[\frac{b_1}{Q_0-1} x^2 + \frac{b_0}{Q_0} x - \frac{1+2Q_0}{1+Q_0} \right] \equiv |x|^{-Q_0-1} R(x).$$

Here, x_g^- and x_g^+ correspond to the x -coordinate of the anti-saddles in the region $x < 0$ and $x > 0$, respectively.

Proof. The standard way to transform a quadratic system without a y^2 -term in the $\frac{dx}{dt}$ equation [14,19] is to introduce a new variable $u = |x|^{-Q_0-1}(1 + xy)$ followed by a rescaling of time $t \rightarrow t |x|^{-Q_0-1}$. Application of this transformation to (4.1) leads to (3.1) with the functions given in (4.6).

Since we consider the case of two foci in each half plane, we assume that $g(x)$ has zeros $x = x_g^+ > 0$ and $x = x_g^- < 0$ corresponding to the foci and that $g(x)$ does not have other real zeros, i.e., the two other zeros of the quartic function $W_4(x)$ are assumed to be complex. The explicit condition to enforce this will not be needed in the following and is omitted.

In each of the two half planes $x < 0$ and $x > 0$, we define the functions $F_-(x)$ and $F_+(x)$, respectively, as $\int_{x_g^-}^x f(\bar{x}) d\bar{x}$ and $\int_{x_g^+}^x f(\bar{x}) d\bar{x}$, respectively. These are the integrals of the divergence function $f(x)$ evaluated from the position of the singularity, i.e., $x = x_g^- < 0$ and $x = x_g^+ > 0$, respectively. $\square$

Note that in the integration it was explicitly used that $Q_0 \in (0, 1)$ avoiding singularities in the final integral expression, i.e., the values $Q_0 = 1, 0, -1$ were avoided. Explicitly the constants in the expressions for $F(x)$ are given by $c_- = -F_0(x_g^-)$ and $c_+ = -F_0(x_g^+)$. However, in the following, we will not need the explicit expressions of the constants. The only information needed in the application of uniqueness theorems is the sign of the constants, which we will deduce from a geometrical argument. We are going to apply Theorems 3.5 and 3.6,

where a constant K was introduced being added to $F_{x_g}(x)$. In our case, K will be directly related to the constants c_- and c_+ through $K = -c_-$ ($K = -c_+$) for $x < 0$ ($x > 0$).

A further reduction of the parameter range can be obtained by observing that the sign of the divergence needs to change in both regions $x < 0$ and $x > 0$ to ensure that a limit cycle distribution is possible.

Lemma 4.9. *A necessary condition for system (4.1) to have a limit cycle distribution (m, n) with $m > 0$, $n > 0$ is $b_1 > 0$.*

Proof. To have a non-trivial (i.e., at least one limit cycle around each focus) distribution of limit cycles the function $W_2(x)$ in (4.6) needs to change sign in both regions $x < 0$ and $x > 0$, because $x = 0$ cannot be crossed by limit cycles according to Lemma 4.6. If $W_2(x)$ were to be of fixed sign in either region, then no limit cycles could exist because the divergence would have fixed sign. This implies that a necessary condition for the existence of a true distribution of limit cycles (m, n) , $b_1 > 0$ needs to hold true. $\square$

4.3 Properties of the Liénard functions

With this information, we can prove the following straightforward properties for the functions $f(x)$, $g(x)$ and $F_0(x)$.

Lemma 4.10. *The function $f(x)$ as defined in (4.6), under the restrictions of Lemmas 4.7 and 4.9 satisfies:*

- (i) $\lim_{x \downarrow 0} f(x) = \infty$, $\lim_{x \uparrow 0} f(x) = -\infty$, $\lim_{x \rightarrow \infty} f(x) \uparrow 0$, $\lim_{x \rightarrow -\infty} f(x) \downarrow 0$,
- (ii) $f(x)$ has two zeros $x_f^- < 0$ and $x_f^+ > 0$,
- (iii) for $-\infty < x < x_f^-$ ($x_f^- < x < 0$) $f(x) > 0$ ($f(x) < 0$) and for $0 < x < x_f^+$ ($x_f^+ < x < \infty$) $f(x) > 0$ ($f(x) < 0$).

Proof. For large x , we obtain from (4.6) as leading terms $f(x) \sim -b_1 x |x|^{-Q_0-1}$. For small x , we obtain from (4.6) as leading terms $f(x) \sim \frac{1+2Q_0}{x} |x|^{-Q_0-1}$. The sign of the functions are determined by the quadratic functions $W_2(x)$ for $f(x)$.

According to the parameter restrictions (4.7) and (4.9), $W_2(x)$ has two zeros $x = x_f^- < 0$ and $x = x_f^+ > 0$. Moreover, $W_2(0) = 1 + 2Q_0 > 0$. $\square$

Lemma 4.11. *The function $g(x)$ as defined in (4.6), under the restrictions of Lemmas 4.7 and 4.9 satisfies:*

- (i) $g(x)$ has two zeros $x_g^- < 0$ and $x_g^+ > 0$,
- (ii) for $-\infty < x < x_g^-$ ($x_g^- < x < 0$) $g(x) < 0$ ($g(x) > 0$) and $0 < x < x_g^+$ ($x_g^+ < x < \infty$) $g(x) < 0$ ($g(x) > 0$).

Proof. The sign of the function $g(x)$ is determined by the quartic function $W_4(x)$. The function $W_4(x)$ has two zeros $x_g^- < 0 < x_g^+$, because we are studying the case with exactly two finite anti-saddles. Moreover, $W_4(0) = -Q_0 < 0$. The statements about the signs of the functions follow from these observations. $\square$

Lemma 4.12. *The function $F_0(x)$ as defined in (4.7), under the restrictions of Lemmas 4.7 and 4.9 satisfies:*

- (i) $\lim_{x \downarrow 0} F_0(x) = -\infty$, $\lim_{x \uparrow 0} F_0(x) = -\infty$, $\lim_{x \rightarrow \infty} F_0(x) = -\infty$, $\lim_{x \rightarrow -\infty} F_0(x) = -\infty$,
- (ii) $F_0(x)$ has a local maximum for $x = x_f^-$ and $x = x_f^+$,
- (iii) $F_0(x)$
 - either has no zeros and then $F_0(x) < 0$ for all $x \neq 0$,
 - or it has two zeros (which could coincide) $x_F^{(1)}$, $x_F^{(2)}$, and then $x_F^{(1)} \cdot x_F^{(2)} > 0$, i.e., they lie in the same region $x < 0$ or $x > 0$. They satisfy $x_F^{(1)} < x_f^- < x_F^{(2)} < 0$ or $0 < x_F^{(1)} < x_f^+ < x_F^{(2)}$ depending on the region where the zeros occur. One limiting case could occur, where $x_F^{(1)} = x_f^- = x_F^{(2)}$, in which case the graph of $y = F_0(x)$ will be tangent to the x -axis.

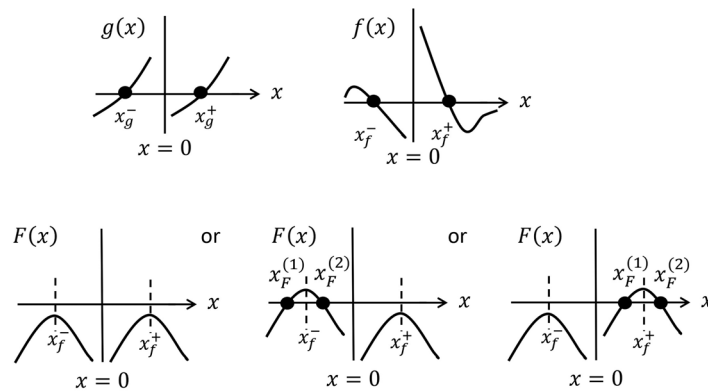


Figure 3: The basic properties of the functions $f(x)$, $g(x)$, and $F_0(x)$ of Lemmas 4.10, 4.11, and 4.12, respectively.

In both cases, $F_0(x) < 0$ in one of the two regions $x < 0$ or $x > 0$.

Proof. For large x , we obtain from (4.7) as leading terms $F_0(x) \sim \frac{-b_1}{1-Q_0} |x|^{1-Q_0}$. For small x , we obtain from (4.7) as leading terms $F_0(x) \sim \frac{-(1+2Q_0)}{1+Q_0} |x|^{1-Q_0}$. The sign of the function is determined by the quadratic function $R(x)$.

The function $R(x)$ could have two zeroes $x_F^{(1)}$, $x_F^{(2)}$. If they exist, then $x_F^{(1)} \cdot x_F^{(2)} > 0$, i.e., they lie in the same region $x < 0$ or $x > 0$. Since $F'(x) = f(x)$ by construction, according to Rolle's theorem between the two zeros $x_F^{(1)}$, $x_F^{(2)}$, a zero of $f(x)$ needs to be positioned, i.e., $x_f^- < x_F^{(1)} < x_f^+ < 0$ or $0 < x_F^{(1)} < x_f^- < x_f^+ < x_F^{(2)}$. $\square$

In Figure 3, the functions are sketched.

4.4 Properties of the function $Z(x)$

The structure of the proof is then to show that in system (3.1) with functions defined by (4.6) and (4.7), we can apply Theorem 3.5 or Theorem 3.6 (i.e., the practical versions of Theorem 3.1 or Theorem 3.3, respectively) in one of the two regions $x < 0$ or $x > 0$, i.e., that we can either prove that no limit cycle exists according to Theorem 3.5 by finding a gap in the function $Z(x) = \frac{F_0}{g}(x)$ (condition (iii)), or at most one limit cycle exists by showing that the conditions (iii) and (iv) are satisfied in Theorem 3.6, i.e., first showing that $Z'(x) < 0$ on an appropriate interval and then showing that (iv) holds true as well.

For (4.6) and (4.7), the system of equation (3.3) in Theorems 3.1 and 3.3 takes the form:

$$\begin{aligned} F_0(x_1) &= F_0(x_2), \\ \frac{f(x_1)}{g(x_1)} &= \frac{f(x_2)}{g(x_2)}, \end{aligned} \quad (4.8)$$

with

$$\begin{aligned} F_0(x) &= |x|^{-Q_0-1} \left(\frac{b_1}{Q_0-1} x^2 + \frac{b_0}{Q_0} x - \frac{1+2Q_0}{1+Q_0} \right) \equiv |x|^{-Q_0-1} R(x), \\ \frac{f}{g}(x) &= |x|^{Q_0+1} \frac{W_2(x)}{W_4(x)}, \\ W_4(x) &\equiv -x^2(a_0 + a_1x - x^2) + x(b_0 + b_1x) - Q_0, \\ W_2(x) &\equiv -b_1x^2 - b_0x + 1 + 2Q_0, \\ R(x) &\equiv \frac{b_1}{Q_0-1} x^2 + \frac{b_0}{Q_0} x - \frac{1+2Q_0}{1+Q_0}. \end{aligned}$$

As noted in Theorems 3.5 and 3.6, the number of solutions can be studied by analyzing the function $Z(x)$ given by:

$$Z(x) \equiv \frac{f(x)}{g(x)} F_0(x) = \frac{R(x)W_2(x)}{W_4(x)} = \frac{\left(\frac{b_1}{Q_0-1}x^2 + \frac{b_0}{Q_0}x - \frac{1+2Q_0}{1+Q_0} \right) (-b_1x^2 - b_0x + 1 + 2Q_0)}{-x^2(a_0 + a_1x - x^2) + x(b_0 + b_1x) - Q_0} \quad (4.9)$$

In the following, we will distinguish two cases for the function $Z(x)$.

Definition 4.13. The following two cases can occur for the functions $g(x)$ and $F_0(x)$ as defined in (4.6) and (4.7), respectively.

Case (A) $g(x)$ and $F_0(x)$ have (at least one) a zero in common.

Case (B) $g(x)$ and $F_0(x)$ do not have a zero in common.

The reason for distinguishing these two cases is that in Case (B), the function $Z(x)$ will have two vertical asymptotes at $x = x_g^-$ and $x = x_g^+$, while in Case (A), it will not have a vertical asymptote for either $x < 0$ or $x > 0$ or both. The application of Theorem 3.5 condition (iii) requires that $F_0(x)$ does not have a common zero with $g(x)$, i.e., for Case (B), it is applicable but not for Case (A). For this reason, we will consider the proof for both cases separately.

In [22], it was merely stated that there are always two asymptotes for the function $Z(x)$. Still there does not seem to be a conflict with the proof in [22], because the existence of two asymptotes was not needed in all cases, and where it was needed, it was proved to be true.

In the next two sections, we will investigate the distribution property for both cases. We will exclude the weak focus case, by enforcing that $x_f^- \neq x_g^-$ and $x_f^+ \neq x_g^+$.

Definition 4.14. We denote by the *strong focus* case the conditions $x_f^- \neq x_g^-$ and $x_f^+ \neq x_g^+$.

The weak focus case will be moved to the section of the non-generic cases.

4.5 Case (A): $g(x)$ and $F_0(x)$ have (at least one) a zero in common

Under the assumption of Case (A) in Definition 4.13, there exists a zero of $F_0(x)$. This means that according to property (iii) of Lemma 4.12, there are three consecutive zeros of $Z(x)$: $x_F^{(1)} < x_f^\pm < x_F^{(2)} < 0$, where $x_f^\pm$ indicates it is either x_f^- or x_f^+ , one of the two zeros of $f(x)$. The three values could coincide $x_F^{(1)} = x_f^\pm = x_F^{(2)}$.

We will prove that in this case in the region where the zeros of $F_0(x)$ do not occur, i.e., the region where $g(x)$ and $F_0(x)$ do not have a common zero, at most one limit cycle exists.

Proposition 4.15. In Case (A) of Definition 4.13 system (3.1) with functions (4.6) and (4.7) under the restrictions of Lemmas 4.7 and 4.9 and the restriction that no singularity is weak (i.e., strong foci only as in Definition 4.14) has limit cycle distributions $(n, 0)$ or $(n, 1)$ with $n \in \{0\} \cup \mathbb{N}$. In particular, at most one limit cycle exists in the region $x < 0$ or $x > 0$, where the functions $g(x)$ and $F_0(x)$ do not have a common zero.

Proof. In Figures 4 and 5, the graphs discussed in the proof are shown.

Assume that the common zero occurs for $x < 0$ (the proof for the case $x > 0$ goes exactly along the same lines). $x_F^{(1)} < x_f^- < x_F^{(2)} < 0$. These three zeros cannot coincide because it would imply the existence of a weak focus.

The function $Z(x)$ is a rational function with polynomial functions of (at most) degree 3 with 3 zeros, $x_f^- < 0$, $x_F^{(1)} < 0$ (or $x_F^{(2)} < 0$), $x_f^+ > 0$ and one vertical asymptote $x = x_g^+ > 0$. Moreover, $Z(0) = \frac{(1+2Q_0)^2}{Q_0(Q_0+1)} > 0$, $Z'(0) = 0$, $\lim_{x \rightarrow \pm\infty} Z(x) \equiv Z_\infty = \frac{b_1^2}{(1-Q_0)} > 0$ (these properties will be true in Case (B) of the next section as well).

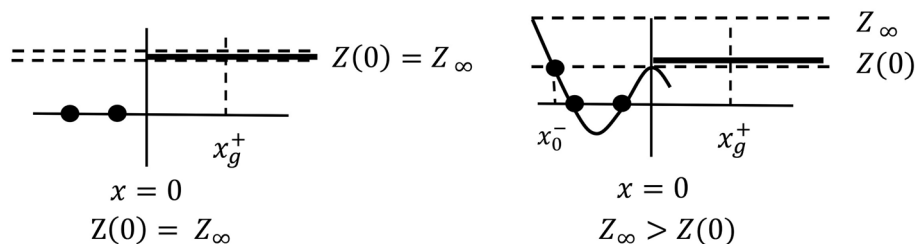


Figure 4: The cases $Z(0) = Z_\infty$ and $Z(0) < Z_\infty$ in the proof of Proposition 4.15. In both cases, there is a gap $y = Z(0)$ in the function $y = Z(x)$ for $x > 0$, and no limit cycles exist.

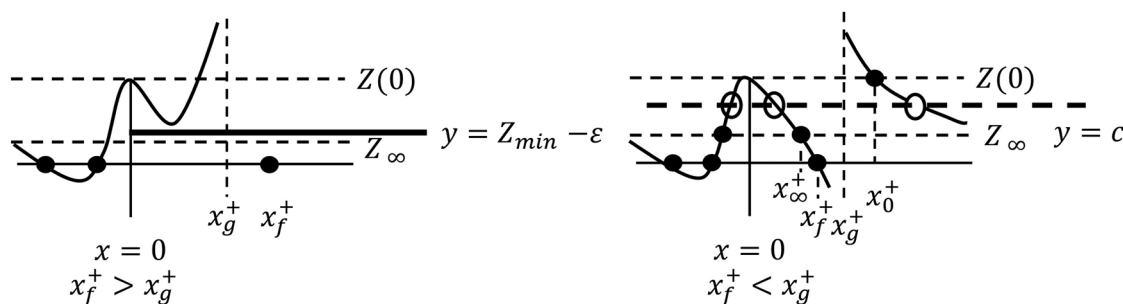


Figure 5: The cases $Z(0) > Z_\infty$ with $x_f^+ > x_g^+$ and $x_f^+ < x_g^+$, respectively, in the proof of Proposition 4.15. If $x_f^+ > x_g^+$, then there is a gap for $x > 0$ at $y = Z_{\min} - \varepsilon$ and no limit cycles exist. If $x_f^+ < x_g^+$, then $\forall c \in (Z_\infty, Z(0))$, there are three intersections with $y = c$ and at most one limit cycle exists.

We distinguish two cases: $Z_\infty < Z(0)$ and $Z_\infty \geq Z(0)$. The intersection of $y = Z(x)$ with horizontal lines $y = c$ is determined by a cubic equation in x . Therefore, at most three such intersections exist for $c = Z(0)$, but since $y = Z(x)$ has quadratic contact with $y = Z(0)$ for $x = 0$ (because $Z'(0) = 0$) only one intersection exists for $x \neq 0$. For $c = Z(\infty)$, only two intersections are possible, because the equation $Z(x) = Z(\infty)$ is only quadratic in x .

If $Z_\infty = Z(0)$, then $Z(x) - Z(\infty)$ does not have any zeroes for $x > 0$, and in Theorem 3.5, condition (iii) holds for $x > 0$ with $c = Z(\infty)$: no limit cycles exist for $x > 0$. See Figure 4.

If $Z_\infty > Z(0)$, then $\lim_{x \rightarrow -\infty} Z(x) = Z_\infty > Z(0)$, and the fact that $Z(x)$ is continuous for all $x < 0$ implies that there exists an $x_0^- < 0$ such that $Z(x_0^-) = Z(0)$. This exhausts all intersections of $Z(x) = Z(0)$, i.e., a double intersection for $x = 0$ and $x_0^- < 0$. It follows that for $x > 0$ $Z(x) - Z(0)$ does not have any zeroes, and in Theorem 3.5, condition (iii) holds for $x > 0$ with $c = Z(0)$: no limit cycles exist for $x > 0$. See Figure 4.

If $Z_\infty < Z(0)$, then we distinguish two situations $x_f^+ > x_g^+$ or $x_f^+ < x_g^+$ (the case $x_f^+ = x_g^+$ corresponds to a weak focus which we do not consider here).

First, we observe that necessarily $Z''(0) < 0$. If $Z''(0) = 0$, then $Z(x) - Z(0)$ would have a triple zero for $x = 0$, meaning that $Z(x) - Z(0)$ would not have any other zeros, i.e., in Theorem 3.5, condition (iii) holds for $x > 0$ with $c = Z(0)$: no limit cycles exist for $x > 0$. If $Z''(0) > 0$, then $Z(-\varepsilon) > Z(0)$ for $0 < \varepsilon \ll 1$. Since $Z(x)$ is continuous for all $x < 0$, and since $Z(x_f^-) = 0$, there must exist a $x_0^- < 0$ such that $Z(x_0^-) = Z(0)$. In a similar way as in the case $Z_\infty > Z(0)$, we then argue that for $x > 0$ $Z(x) - Z(0)$ does not have any zeroes, and in Theorem 3.5, condition (iii) holds for $x > 0$ with $c = Z(0)$: no limit cycles exist for $x > 0$.

Therefore, we only need to consider the case $Z''(0) < 0$ next.

If $x_f^+ > x_g^+$, then $Z(x) > 0$ for $0 < x < x_g^+$. Since $\lim_{x \uparrow x_g^+} Z(x) = \infty$ and $Z''(0) < 0$ implies that $Z(\varepsilon) < Z(0)$ for $0 < \varepsilon \ll 1$, there must exist a x_0^+ with $0 < x_0^+ < x_g^+$ such that $Z(x_0^+) = Z(0)$. Rolle's theorem then says that there exists a $x_{\min}$ with $0 < x_{\min} < x_0^+$ such that $Z'(x_{\min}) = 0$, where $Z(x)$ has a positive local minimum $0 < Z(x_{\min}) < Z(0)$.

For $x < 0$, the continuous function $Z(x)$ becomes zero for x_f^- . Therefore, there is a $x^* < 0$ such that $Z(x) = Z(x_{\min})$. We obtain that $Z(x) - Z(x_{\min})$ has a double zero for $x = x_{\min}$ and a simple zero for $x^* < 0$ (because $Z(x) - Z(x_{\min}) = 0$ leads to a cubic equation in x). Then $Z(x) - Z(x_{\min}) + \varepsilon_1$, $0 < \varepsilon_1 \ll 1$, i.e., shifting the graph of $y = Z(x)$ slightly downwards gives a function which is positive for all $x > 0$. Therefore, for $x > 0$ $Z(x) - Z(x_{\min}) + \varepsilon_1$ does not have any zeros, and in Theorem 3.5, condition (iii) holds for $x > 0$ with $c = Z(x_{\min}) - \varepsilon_1$: no limit cycles exist for $x > 0$. See Figure 5.

If $x_f^+ < x_g^+$, then continuity of $Z(x)$ for $0 < x < x_f^+$, with $Z(0) > Z_\infty$, $Z(x_f^+) = 0$, implies that there exists x_∞ with $0 < x_\infty^+ < x_f^+$ such that $Z(x_\infty^+) = Z_\infty$. In the same way as for the case $x_f^+ > x_g^+$, it can be argued that continuity of $Z(x)$ for all $x < 0$ implies that there exists a $x_\infty^- < 0$ such that $Z(x_\infty^-) = Z_\infty$. The implication is that the horizontal asymptote $y = Z_\infty$ is not crossed at any other value of x .

For $x > x_g^+$, $y = Z(x)$ needs to cross $y = Z(0)$. If not, then taking $c = F(0)$ in Theorem 3.5 condition (iii) would show that there are no limit cycles. We denote this unique intersection by x_0^+ . Since we just proved that $y = Z_\infty$ is not crossed for any $x > 0$ except for $0 < x_\infty^+ < x_f^+$, it follows that for $x > x_0^+$, $Z_\infty < Z(x) < Z(0)$.

The function $Z(x)$ then has the following structure for $x > 0$:

$$\begin{aligned} 0 < x < x_\infty^+, & \quad Z_\infty < Z(x) < Z(0), \\ x_\infty^+ < x < x_g^+, & \quad Z(x) < Z_\infty, \\ x_g^+ < x < x_0^+, & \quad Z(x) > Z(0), \\ x > x_0^+, & \quad Z_\infty < Z(x) < Z(0). \end{aligned}$$

From this, it follows that the equation $Z(x_1) = Z(x_2)$ with $0 < x_1 < x_g^+ < x_2$ only has solutions for $0 < x_1 < x_f^+$, $x_2 > x_0^+$. We apply Theorem 3.5 condition (iii) with this interval for x where we need to prove that $Z'(x)$ has fixed sign. Since $Z(x)$ is continuous for $x < 0$, we find that for every $c \in (Z_\infty, Z(0))$, there is a corresponding $x < 0$ such that $Z(x) = c$. It implies that for every $c \in (Z_\infty, Z(0))$, $Z(x) = c$ has three solutions. This is the maximum number of solutions because $Z(x) = c$ is a cubic equation in x . It follows that $Z'(x)$ cannot be zero for any x in $0 < x < x_f^+$, $x > x_0^+$. In particular, in this case $Z'(x) < 0$, showing that condition (iii) is satisfied. See Figure 5.

It remains to prove that condition (iv) is satisfied. We rewrite the condition in the following form. The function $\frac{f(x)F_{xg}(x)}{g(x)}$ can be rewritten in terms of $Z(x)$, using the fact that $Z(x) = \frac{f(x)F_0(x)}{g(x)}$, from which we deduce that $\frac{f(x)}{g(x)} = \frac{Z(x)}{F_0(x)}$. Substitution gives $\frac{f(x)F_{xg}(x)}{g(x)} = \frac{Z(x)F_{xg}(x)}{F_0(x)}$. The sign of the derivative of this function (which is needed in condition (iii) of Theorem 3.6) is then determined by $f(x)Z(x)K + Z'(x)F(x)F_0(x)$. To find K , we observe that $F_{xg}(x_f^+) + K = 0 + K = F_0(x_f^+) < 0$, i.e., $K < 0$. On the interval $0 < x < x_f^+$, we have $f(x) > 0$, $Z(x) > 0$, $K < 0$, $Z'(x) < 0$, $F(x) < 0$, $F_0(x) < 0$, i.e., $f(x)Z(x)K + Z'(x)F(x)F_0(x) < 0$. The first case of condition (iv) applies, proving that at most one limit cycle exists. $\square$

It would be natural to assume that this proof can be used for the weak focus case in system (3.1) with functions (4.6) and (4.7) as well. If $f(x)$ and $g(x)$ have a factor in common, i.e., the function $Z(x)$ would lose a vertical asymptote. However, the aforementioned proof would not work, because an essential element was that for $x < 0$ $Z(x)$ still had two zeros. This might not be true for a weak focus, e.g. in the situation when $F_0(x)$ does not have zeros at all.

This idea of the proof in Case (A) extends in a natural way to Case (B) of the following section. The arguments are the same, with the main difference being that the function $Z(x)$ in Case (B) is a rational function with quartic polynomials. However, the main argument is still based on scanning the graph of $y = Z(x)$ with horizontal lines and using the fact that intersections correspond to zeros of a quartic function.

4.6 Case (B): $g(x)$ and $F_0(x)$ do not have a zero in common

The following properties for the function $Z(x)$ hold in Case (B).

Lemma 4.16. The function $Z(x)$ as defined in (4.9) under the restrictions of Lemmas 4.7 and 4.9 and the assumption that $g(x)$ and $F_0(x)$ do not have a common zero (Case (B) of Definition 4.13), and the assumption that $f(x)$ and $g(x)$ do not have a common zero (Definition 4.14) satisfies:

- (1) $\lim_{x \rightarrow \pm\infty} Z(x) \equiv Z_\infty = \frac{b_1^2}{(1-Q_0)} > 0$.
- (2) $Z(0) = \frac{(1+2Q_0)^2}{Q_0(Q_0+1)} > 0$.
- (3) $Z(x)$ has (at least) two zeroes for $x = x_f^- < 0$, $x = x_f^+ > 0$, for which $W_2(x) = 0$.
- (4) $Z(x)$ could have two more zeroes $x_F^{(1)}, x_F^{(2)}$, for which $R(x) = 0$. If $x_F^{(1)}, x_F^{(2)}$ exist, then $x_F^{(1)} x_F^{(2)} > 0$, i.e., they lie in the same half plane $x < 0$ or $x > 0$. Moreover, $x_F^{(1)} < x_f^+ < x_F^{(2)}$ or $x_F^{(1)} = x_f^+ = x_F^{(2)}$.
- (5) The graph of $y = Z(x)$ has exactly two vertical asymptotes $x = x_g^- < 0$ and $x = x_g^+ > 0$.
- (6) The graph of $y = Z(x)$ intersects every horizontal line $y = c$ in at most four points.
- (7) $Z'(0) = 0$.
- (8) The graph of $y = Z(x)$ is tangent to the line $y = Z(0)$. It intersects the line $y = Z(0)$ in at most two other points for $x \neq 0$.
- (9) The graph of $y = Z(x)$ intersects the horizontal asymptote $y = Z_\infty$ in at most three points.

Proof. Properties (1) and (2) follow directly from the definition of the functions $Z(x)$ and the restrictions of Lemmas 4.7 and 4.9, i.e., $0 < Q_0 < 1$ and $b_1 > 0$.

Properties (3), (4), and (5) follow Lemmas 4.12, 4.11, and 4.10, respectively.

Property (6) follows from the fact that the equation $Z(x) = c$ is a quartic function in x because both numerator and denominator of $Z(x)$ are quartic.

Property (7) is the critical observation that was made in [22]. It follows from a straightforward calculation of the derivative of $Z(x)$ as defined in (4.9).

Property (8) is a consequence of property (7): the tangent line at $x = 0$ to $y = Z(x)$ is horizontal. Since according to property (6) the equation $Z(x) = Z(0)$ has at most four solutions, and since $y = Z(x)$ has at least order two contact at $x = 0$ with its tangent line, the statement follows.

Property (9) follows from the fact the equation $Z(x) = Z_\infty$ is cubic in x . □

In Figure 6, some of the properties of the lemma are shown.

We will analyze $Z(x)$ and show the following: either $Z'(x)$ has fixed sign on a relevant interval for x or the function $Z(x)$ has a gap, i.e., there is a $c \in \mathbb{R}$ such that $Z(x) = c$ does not have a solution. In the former case, the system (4.8) will have at most one solution pair (x_1, x_2) , while in the latter case, it will have no solutions.

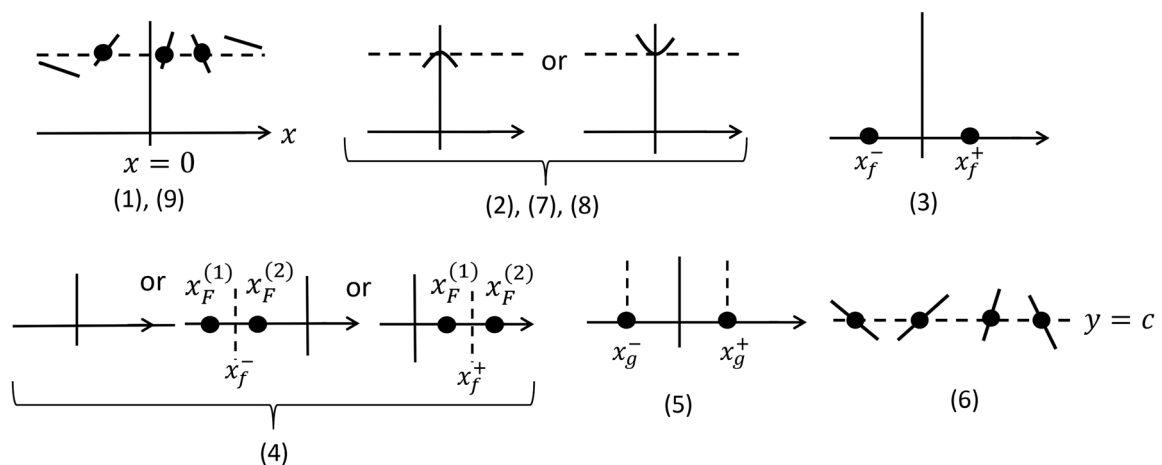


Figure 6: The properties of the functions $Z(x)$ of Lemmas 4.16.

To study $Z(x)$, we distinguish different cases based on the location of the intersections of $y = Z(x)$ with $y = Z(0)$. According to property (8) of Lemma 4.16, the curve of $y = Z(x)$ has at most two intersections with the horizontal line $y = Z(0)$ for $x \neq 0$. First, we make the observation that these two intersections necessarily need to exist to have two non-empty nests of limit cycles.

Lemma 4.17. *In Case (B) of Definition 4.13 a necessary condition for the Liénard system (3.1) with functions (4.6) and (4.7) under the restrictions of Lemmas 4.7 and 4.9 and the restriction that no singularity is weak (i.e., strong foci only as in Definition 4.14) to have two nests of limit cycles, i.e., a distribution (n, m) with $n > 0, m > 0$, is that $Z(x) = Z(0)$ as defined in (4.9) has two solutions $x \neq 0$, i.e., $x_0^- < 0, x_0^+ > 0$.*

Proof. Suppose that the equation $Z(x) = Z(0)$ does not have a solution for $x > 0$. According to Theorem 3.5, there is no limit cycle for $x > 0$. The reason is that there are no common factors between $F_0(x)$, $f(x)$, and $g(x)$, and $Z(x) - Z(0)$ has fixed sign. Therefore, to have limit cycles for $x > 0$, there must exist a $x_0^+ > 0$ such that $Z(x_0^+) = Z(0)$. The same argument applies to the region $x < 0$ proving the existence of $x_0^- < 0$ such that $Z(x_0^-) = Z(0)$. $\square$

With this in mind, we can categorize different cases for the function $Z(x)$ depending on the relative positions of x_0^- , x_g^- for $x < 0$ and x_0^+ , x_g^+ for $x > 0$. Moreover, in each of these cases, we need to distinguish two additional cases: whether $Z(0) < Z_\infty$ or $Z(0) > Z_\infty$. Note that the case $Z(0) = Z_\infty$ cannot occur due to Properties (8) and (9) in Lemma 4.16: there is (at least) double contact between $y = Z(x)$ and $y = Z(0)$ at $x = 0$ according to Property (8), whereas at most three intersections in total can occur according to Property (9) which would contradict the existence of two additional intersections $x_0^- < 0, x_0^+ > 0$ as stated in Lemma 4.17.

The sign of $Z''(0)$ is fixed according to whether $Z(0)$ is less than or larger than Z_∞ . This will restrict the number of cases to study:

Lemma 4.18. *In Case (B) of Definition 4.13 if the Liénard system (3.1) with functions (4.6) and (4.7) under the restrictions of Lemmas 4.7 and 4.9 and the restriction that no singularity is weak (i.e., strong foci only as in Definition 4.14) has two nests of limit cycles, i.e., a distribution (n, m) with $n > 0, m > 0$, then $Z''(0) < 0$ ($Z''(0) > 0$) if $Z(0) > Z_\infty$ ($Z(0) < Z_\infty$) as defined in (4.9).*

Proof. Consider the case $Z(0) > Z_\infty$. Suppose that $Z''(0) > 0$. For $x > 0$, the function $Z(x)$ is continuous except for $x = x_g^+$ where it has a vertical asymptote. Formally, we can write $Z(x) - Z(0) = \frac{p(x)}{x - x_g^+}$, where $p(x)$ is continuous for all $x > 0$. For small positive x , $Z(x) - Z(0) > 0$ by virtue of the assumption that $Z''(0) > 0$ and the fact that $Z'(0) = 0$ according to Property (7) of Lemma 4.16. For large enough x , we have $Z(x) - Z(0) < 0$ since $Z(0) > Z_\infty$. It means that on the interval $0 < x < \infty$, the function $Z(x) - Z(0)$ changes sign an odd number of times. Crossing the discontinuity at $x = x_g^+$ changes the sign of $Z(x) - Z(0)$, which means that effectively on the interval $0 < x < \infty$ the function has an even number of zeros. However, that contradicts the statement of Lemma 4.17 that there is a unique $x_0^+ > 0$ such that $Z(x_0^+) = Z(0)$. Therefore, the assumption that $Z''(0) > 0$ is false. See Figure 7.

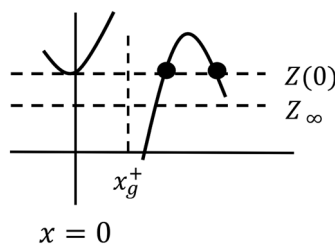


Figure 7: Contradiction for the case when $Z(0) > Z_\infty$ and $Z''(0) > 0$ in the proof of Lemma 4.18.

In exactly the same way, it can be proved that if $Z(0) < Z_\infty$, the assumption $Z''(0) < 0$ will lead to a contradiction. $\square$

With Lemma 4.18 in mind, the following cases remain to be considered.

Lemma 4.19. *In Case (B) of Definition 4.13 to study the nests of limit cycles in Liénard system (3.1) with functions (4.6) and (4.7) under the restrictions of Lemmas 4.7 and 4.9 and the restriction that no singularity is weak (i.e., strong foci only as in Definition 4.14), the following eight cases need to be considered with $Z(x)$ as defined in (4.9):*

- (1) $x_g^- < x_0^- < 0, 0 < x_g^+ < x_0^+, Z(0) > Z_\infty, Z''(0) < 0$.
- (2) $x_g^- < x_0^- < 0, 0 < x_0^+ < x_g^+, Z(0) > Z_\infty, Z''(0) < 0$.
- (3) $x_0^- < x_g^- < 0, 0 < x_g^+ < x_0^+, Z(0) > Z_\infty, Z''(0) < 0$.
- (4) $x_0^- < x_g^- < 0, 0 < x_0^+ < x_g^+, Z(0) > Z_\infty, Z''(0) < 0$.
- (5) $x_g^- < x_0^- < 0, 0 < x_g^+ < x_0^+, Z(0) < Z_\infty, Z''(0) > 0$.
- (6) $x_g^- < x_0^- < 0, 0 < x_0^+ < x_g^+, Z(0) < Z_\infty, Z''(0) > 0$.
- (7) $x_0^- < x_g^- < 0, 0 < x_g^+ < x_0^+, Z(0) < Z_\infty, Z''(0) > 0$.
- (8) $x_0^- < x_g^- < 0, 0 < x_0^+ < x_g^+, Z(0) < Z_\infty, Z''(0) > 0$.

Here, x_g^- and x_g^+ correspond to the two real zeros of $W_4(x)$ as defined in (4.8), and x_0^- and x_0^+ correspond to the two non-zero solutions to $Z(x) = Z(0)$ following Lemma 4.17.

In Figure 8, the eight cases are shown.

It remains to determine for each case whether for $x < 0$ or $x > 0$, the function $Z(x)$ either has a gap (Theorem 3.5) or is monotonic in an appropriate interval (Theorem 3.6).

Of the eight cases in Lemma 4.19, two can be eliminated by noting that they are essentially the same as two others.

Lemma 4.20. *Cases (1) and (4) in Lemma 4.19 are the same.*

Cases (5) and (8) in Lemma 4.19 are the same.

Proof. Through the transformation $x \rightarrow -x, y \rightarrow -y$, the Liénard system (3.1) transforms into a new Liénard system with $g_{\text{new}}(x) = g(-x), f_{\text{new}}(x) = f(-x)$. The cases of Lemma 4.19 do not depend a priori on the location of the zeroes of $f(x)$. The transformation switches the position of the singularities for $x < 0$ and $x > 0$.

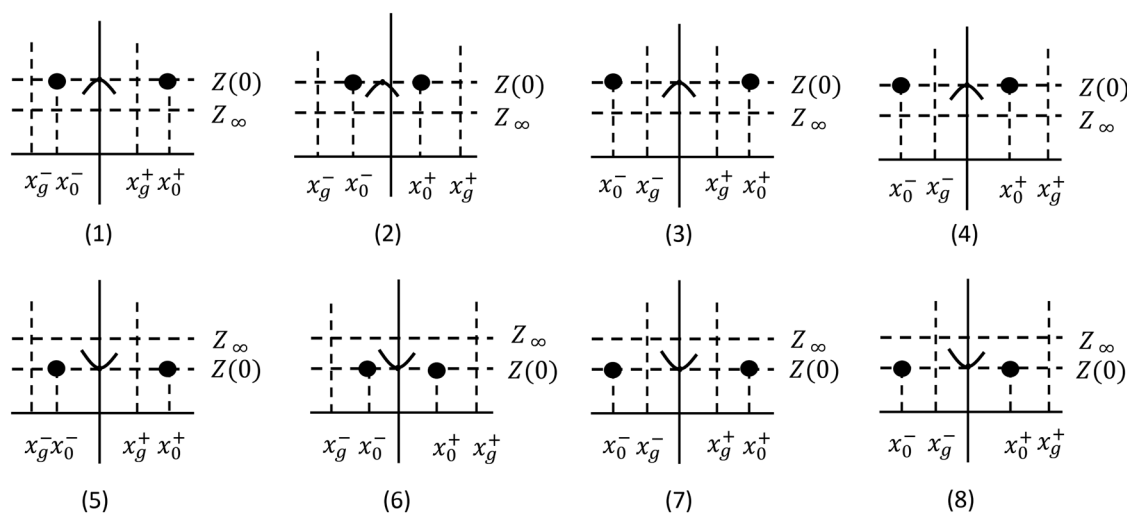


Figure 8: The eight cases for $Z(x)$ of Lemma 4.19.

However, this essentially switches the position of x_g^- and x_g^+ from Case (1) to Case (4) and similarly for the Cases (5) and (8). The relative position of these zeroes with respect to x_0^- and x_0^+ is not influenced by the transformation and neither is the sign of $Z''(0)$. $\square$

It remains to consider the Cases (1), (2), (3), (5), (6), and (7), of Lemma 4.19.

Lemma 4.21. *Case (1) in Lemma 4.19. For the function $Z(x)$ as defined in (4.9) under the strong focus condition (4.14), there are two possibilities:*

- (1) *either $\exists c \in \mathbb{R}^+$ such that $Z(x) - c$ has fixed sign for $x < 0$,*
- (2) *or for $x > 0$ the equation $Z(x_1) = Z(x_2)$ only has solutions for $0 < x_1 < \bar{x}_1 < x_g^+$, $x_g^+ < \bar{x}_2 < x_2$, with $\bar{x}_1 = x_\infty^+$ with $Z(x_\infty^+) = Z_\infty$, $\bar{x}_2 = x_0^+$. On the intervals $0 < x < \bar{x}_1$ and $\bar{x}_2 < x$, $Z'(x) < 0$ holds true.*

Proof. The idea behind the proof is to construct the graph of $y = Z(x)$ step by step (similar to the proof of Proposition 4.15) and identify the two cases in the lemma. The different steps will be illustrated with in Figures 9 and 10 because graphically the result of the lemma is easier to understand.

Step (1) On the interval $0 < x < x_g^+$, $Z(x) < Z(0)$, because $Z(x)$ is continuous, $Z(x) < Z(0)$ for small positive x due to $Z'(0) = 0$, $Z''(0) < 0$ and $Z(x) \neq Z(0)$ by assumption (the only value of $x > 0$ for which $Z(x) = Z(0)$ is $x_0^+ > x_g^+$). Since $Z(x)$ has a vertical asymptote for $x = x_g^+$ necessarily $\lim_{x \uparrow x_g^+} Z(x) = -\infty$. In particular, it means that there is a $0 < x_\infty^+ < x_g^+$ such that $Z(x_\infty^+) = Z_\infty$. According to Property (9) of Lemma (4.16) $Z(x) = Z_\infty$ has at most three solutions.

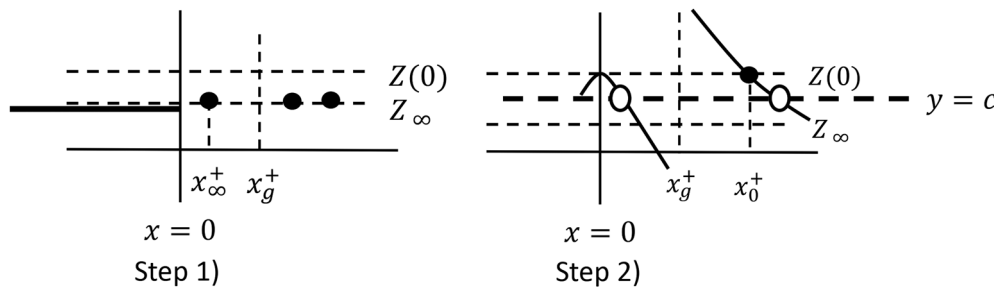


Figure 9: Illustration of the steps (1) and (2) in the proof of Lemma 4.21. Step (1) If there are three positive intersections with the horizontal asymptote, then no limit cycles exist for $x < 0$. Step (2) If there is one positive intersection with the horizontal asymptote, then there are two positive intersections $Z(x) = c$, $\forall c \in (Z_\infty, Z(0))$.

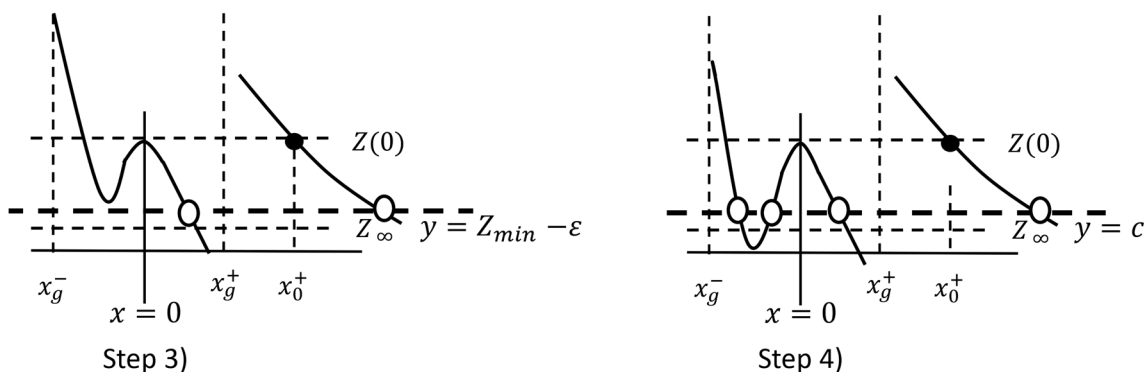


Figure 10: Illustration of the steps (3) and (4) in the proof of Lemma 4.21. Step (3): If $Z_{\min} > Z_\infty$, then for $x < 0$ there is a gap $y = Z_{\min} - \epsilon$ and no limit cycles exist for $x < 0$. Step (4): If $Z_{\min} \leq Z_\infty$, $\forall c \in (Z_\infty, Z(0))$, there are four intersections $Z(x) = c$, $\forall c \in (Z_\infty, Z(0))$ and at most one limit cycle exists for $x > 0$.

Suppose there is more than one x_∞^+ . Then necessarily there are three because $Z(x)$ tends to $-\infty$ when x approaches x_g^+ . It follows that in this case $Z(x) = Z_\infty$ does not have solutions $x < 0$, i.e., choosing $c = Z_\infty$ gives the first case of the lemma. See Figure 9.

Suppose next that x_∞^+ is unique. Since $\lim_{x \downarrow x_g^+} Z(x) = -\infty$, we have $\lim_{x \downarrow x_g^+} Z(x) = \infty$, i.e., the graph of $y = Z(x)$ is decreasing from ∞ for $x = x_g^+ + \varepsilon$ with increasing positive ε . The graph will approach and cross the line $y = Z(0)$ at $x = x_0^+$, meaning that the range of the function $Z(x)$ is $(Z_\infty, Z(0))$. Then it will continue to approach $y = Z_\infty$ for large x , meaning that the range of the function $Z(x)$ is (at least) $(Z_\infty, Z(0))$ for $x > x_0^+$. The implication is that we have established that the graph of $y = Z(x)$ will intersect the horizontal line $y = c$ with $c \in (Z_\infty, Z(0))$ at least twice.

Step (2) On the interval $x_0^- < x < 0$ ($x_g^- < x < x_0^-$), $Z(x) < Z(0)$ ($Z(x) > Z(0)$), because $Z(x)$ is continuous, $Z(x) < Z(0)$ for small negative x due to $Z'(0) = 0$, $Z''(0) < 0$. Since $Z(x)$ has a vertical asymptote for $x = x_g^-$ necessarily $\lim_{x \downarrow x_g^-} Z(x) = \infty$. It follows that $Z(x)$ has (at least) a minimum for $x_{\min}$ with $x_g^- < x_{\min} < 0$. We denote that minimum by $Z_{\min} = Z(x_{\min})$, where $Z_{\min} < Z(0)$.

There are two cases: $Z_{\min} > Z_\infty$ or $Z_{\min} \leq Z_\infty$.

Step (3) If $Z_{\min} > Z_\infty$, then consider the horizontal line $y = Z_{\min}$. We had already established that for $x > 0$ the horizontal line $y = Z_{\min}$ will cross $y = Z(x)$ at least twice, because $Z_\infty < Z_{\min} < Z(0)$. Since $y = Z_{\min}$ has (at least) double contact with $y = Z(x)$ at $x_{\min}$ and since $Z(x) = c$ is a quartic equation in x , we obtain that $Z(x) - Z_{\min}$ has two simple zeros for $x > 0$ and one double zero for $x = x_{\min}$. Take $c = Z_{\min} - \varepsilon$ with $0 < \varepsilon \ll 1$, then $Z(x) - c$ has no solutions for $x < 0$, i.e., Case (1) of the lemma occurs. See Figure 10.

Step (4) If $Z_{\min} \leq Z_\infty$, then any horizontal line $y = c$ with $Z_\infty < c < Z(0)$ will have (at least) two intersection for $x < 0$ with $y = Z(x)$ because of the continuity of $Z(x)$ for $x_g^- < x < 0$. It follows with the two intersections which we already had established for $x > 0$ that $Z(x) - c$ has exactly four simple zeros with the two positive zeroes x_1, x_2 satisfying $0 < x_1 < x_\infty^+$ and $x_2 > x_0^+$. Since they are simple zeros, necessarily for each value of x in these intervals $Z'(x) \neq 0$, which in this case is easily checked to be negative: $Z'(x) < 0$.

It remains to prove that there are no other pairs $0 < x_1 < x_g^+, x_g^+ < x_2$ such that $Z(x_1) = Z(x_2)$ has a solution. Since in this case $Z_{\min} \leq Z_\infty$, there are two values $x_\infty^{(1)}$ and $x_\infty^{(2)}$ with $x_g^- < x_\infty^{(1)} < x_\infty^{(2)} < 0$ such that $Z(x) = Z_\infty$. We had already established that there is a $0 < x_\infty^+ < x_g^+$ such that $Z(x_\infty^+) = Z_\infty$, which is the maximum possible, and that for $x > x_g^+$ the graph of $y = Z(x)$ cannot cross it, i.e., $Z(x) > Z_\infty$. It follows that for $x \in (x_\infty^+, x_g^+)$ $Z(x) < Z_\infty$ and for $x > x_g^+$ $Z(x) > Z_\infty$, i.e., $Z(x_1) = Z(x_2)$ cannot have solutions for this range of x -values. See Figure 10.

This exhausts all possibilities and the result of the lemma follows. $\square$

Lemma 4.22. Case (2) in Lemma 4.19. For the function $Z(x)$ as defined in (4.9) under the strong focus condition (4.14) $\exists c \in \mathbb{R}$ such that $Z(x) - c$ has fixed sign for $x < 0$ or $x > 0$.

Proof. For illustration, the two steps of the proof are shown in Figure 11.

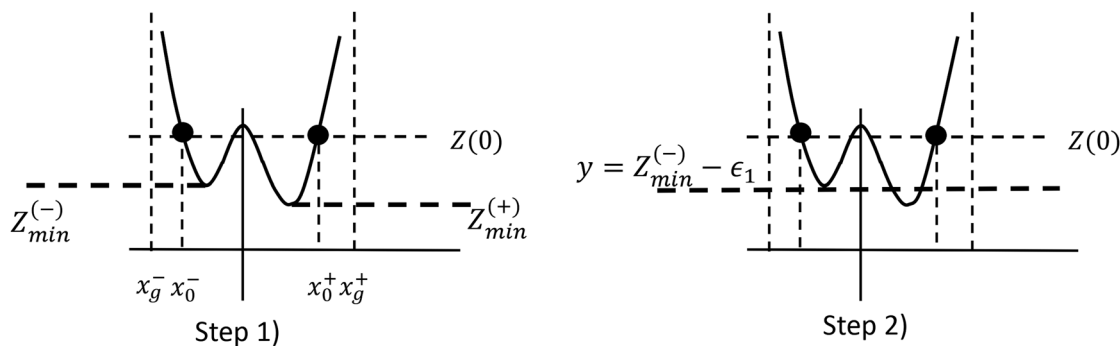


Figure 11: The two steps in the proof of Lemma 4.22. Step (1) Existence of local minima of $Z(x)$. Step (2) Detection of a gap in the graph of $y = Z(x)$. Here, the case $x < 0$ is shown, i.e., $Z_{\min}^0 = Z_{\min}$ is the largest minimum and a gap occurs for $x < 0$.

Step (1) In the same way as in the proof for Lemma 4.21, we find that on the interval $x_0^- < x < 0$ ($x_g^- < x < x_0^-$, $Z(x) < Z(0)$ ($Z(x) > Z(0)$), because $Z(x)$ is continuous, $Z(x) < Z(0)$ for small negative x due to $Z'(0) = 0$, $Z''(0) < 0$. Since $Z(x)$ has a vertical asymptote for $x = x_g^-$ necessarily $\lim_{x \downarrow x_g^-} Z(x) = \infty$. It follows that $Z(x)$ has (at least) a minimum for $x_{\min}^{(-)}$ with $x_g^- < x_{\min}^{(-)} < 0$. We denote that minimum by $Z_{\min}^{(-)} = Z(x_{\min}^{(-)})$, where $Z_{\min}^{(-)} < Z(0)$.

In this case, the same situation occurs in the region $x > 0$, because now we have $0 < x_0^+ < x_g^+$. It follows that there exists $x_{\min}^{(+)}$ with $0 < x_{\min}^{(+)} < x_g^+$. We denote that minimum by $Z_{\min}^{(+)} = Z(x_{\min}^{(+)})$, where $Z_{\min}^{(+)} < Z(0)$.

Step (2) Take the largest local minimum $Z_{\min} = \max[Z_{\min}^{(-)}, Z_{\min}^{(+)}]$, and label its x -value as $x_{\min}$. Denote the other local minimum by $\bar{x}_{\min}$, i.e., $Z(x_{\min}) \geq Z(\bar{x}_{\min})$. By construction, the line $y = Z_{\min}$ is tangent to $y = Z(x)$ at $x = x_{\min}$ and it will intersect $y = Z(x)$ in two points $x_{(1)}$ and $x_{(2)}$ in the region where $\bar{x}_{\min}$ is positioned. These two points exist (they could coincide and form a double intersection), because $\lim_{x \downarrow x_g^-} Z(x) = \lim_{x \uparrow x_g^+} Z(x) = \infty$. In this way, we have constructed a situation where $Z(x) - Z(x_{\min})$ has two simple zeroes (or a double zero) and one double zero. Then $Z(x) - Z(x_{\min}) + \varepsilon_1$, $0 < \varepsilon_1 \ll 1$, i.e., shifting the graph of $y = Z(x)$ slightly downwards gives a function which is positive for all x in the region Ω (which is either $x > 0$ or $x < 0$) where $x_{\min}$ is lying. Therefore, $Z(x) - Z(x_{\min}) + \varepsilon_1$ does not have any zeros in Ω , and in Theorem 3.5, condition (iii) holds for $x \in \Omega$ with $c = Z(x_{\min}) - \varepsilon_1$: no limit cycles exist for $x \in \Omega$. $\square$

Lemma 4.23. Case (3) in Lemma 4.19. For the function $Z(x)$ as defined in (4.9) under the strong focus condition (4.14)

- (a) either for $x < 0$, the equation $Z(x_1) = Z(x_2)$ only has solutions for $x_1 < \bar{x}_1 < x_g^-$, $x_g^- < \bar{x}_2 < x_2 < 0$, with $\bar{x}_1 = x_0^-$, $\bar{x}_2 = x_{\infty}^-$ with $Z(x_{\infty}^-) = Z_{\infty}$. On the intervals $x < \bar{x}_1$, $\bar{x}_2 < x < 0$, $Z'(x) > 0$ holds true,
- (b) or for $x > 0$, the equation $Z(x_1) = Z(x_2)$ only has solutions for $0 < x_1 < \bar{x}_1 < x_g^+$, $x_g^+ < \bar{x}_2 < x_2$, with $\bar{x}_1 = x_{\infty}^+$ with $Z(x_{\infty}^+) = Z_{\infty}$, $\bar{x}_2 = x_0^+$. On the intervals $0 < x < \bar{x}_1$, $\bar{x}_2 < x$, $Z'(x) < 0$ holds true.

Proof. For illustration, the three steps of the proof are shown in Figure 12.

Step (1) On the interval $x_g^- < x < x_g^+$ the function $Z(x) < Z(0)$. Since $\lim_{x \downarrow x_g^-} Z(x) = \lim_{x \uparrow x_g^+} Z(x)$ is infinite, it has to be negative because $y = Z(0)$ cannot be crossed and near $x = 0$ we have $Z(\varepsilon) < Z(0)$, with $0 < |\varepsilon| \ll 1$, i.e., $\lim_{x \downarrow x_g^-} Z(x) = \lim_{x \uparrow x_g^+} Z(x) = -\infty$. It follows that $Z(x)$ has a (at least one) zero $x_g^- < x_z^- < 0$ and a (at least one) zero $0 < x_z^+ < x_g^+$. Therefore, $y = Z(x)$ will cross $y = Z_{\infty} < Z(0)$ an odd number of times on $x_g^- < x < 0$ and an odd number of times on $0 < x < x_g^+$. Since the total number of crossing is less than three in total according to property (9) in Lemma 4.16, the number of crossings has to be exactly one for $x_g^- < x_{\infty}^- < 0$ and exactly one for $0 < x_{\infty}^+ < x_g^+$.

Step (2) Due to property (9) in Lemma 4.16 $y = Z(x)$ can cross $y = Z_{\infty}$ only once more. Suppose this crossing appears for $x < 0$ (if there is not additional crossing the proof remains the same and we will have at most a (1, 1) distribution), at $x = x_{\infty}^*$ with $x_{\infty}^* < x_0^- < x_g^-$. Here, we used that $\lim_{x \downarrow x_g^-} Z(x) = \infty$ and therefore necessarily $x_{\infty}^* < x_0^-$. If this were not the case, there would be additional crossings of the line $Z = Z(0)$ which is prohibited under the conditions of the lemma.

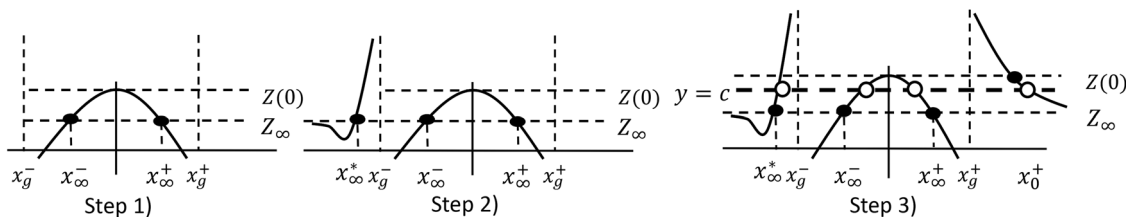


Figure 12: The three steps in the proof of Lemma 4.23. Step (1): Existence of two intersections with $y = Z_{\infty}$. Step (2): The proof that $x_{\infty}^* < x_0^-$. Step (3): For $c \in (Z_{\infty}, Z(0))$, $y = c$ has exactly four intersections with $y = Z(x)$.

Step (3) We have established the following ranges for the function $Z(x)$:

- for $x < x_\infty^*$, $Z(x) < Z_\infty$,
- for $x_\infty^* < x < x_0^-$, $Z_\infty < Z(x) < Z(0)$,
- for $x_0^- < x < x_g^-$, $Z(0) < Z(x) < \infty$,
- for $x_\infty^- < x < 0$, $Z_\infty < Z(x) < Z(0)$,
- for $0 < x < x_\infty^+$, $Z_\infty < Z(x) < Z(0)$,
- for $x_\infty^+ < x < x_g^+$, $Z(x) < Z_\infty$,
- for $x_g^+ < x < x_0^+$, $Z(x) > Z(0)$,
- for $x_0^+ < x$, $Z_\infty < Z(x) < Z(0)$.

We see that in the region $x > 0$ the equation $Z(x_1) = Z(x_2)$ can only have solutions $0 < x_1 < x_g^+ < x_2$ for $0 < x_1 < x_\infty^+$ and $x_0^+ < x_2$. On those intervals, the range of $Z(x)$ is $Z_\infty < Z(x) < Z(0)$. However, we found for $x_\infty^* < x < x_0^-$ and $x_\infty^- < x < 0$ the same range of $Z(x)$, i.e., every line $y = c$ with $c \in (Z_\infty, Z(0))$ has exactly four intersections with $y = Z(x)$. It follows that $Z(x) - c$ has exactly four simple zeros because the equation $Z(x) = c$ is quartic (property (6) in Lemma 4.16). Therefore, $Z'(x)$ has fixed sign in the intervals $0 < x < x_\infty^+$ and $x_0^+ < x$, i.e., $Z'(x) < 0$ as we set out to prove for case (2) of the lemma.

If in step (2) the crossing of $y = Z_\infty$ would occur for $x > 0$, exactly the same line of reasoning would prove that in the region $x < 0$ $Z'(x) > 0$ for the relevant intervals of x . This would lead to case (1) in the lemma. $\square$

Lemma 4.24. Case (4) in Lemma 4.19 can be reduced to Case (1) in Lemma 4.21 according to Lemma 4.20.

Lemma 4.25. Cases (5), (6), (7), (8) in Lemma 4.19.

Case (5) For the function $Z(x)$ as defined in (4.9) under the strong focus condition (4.14), there are two possibilities:

- either $\exists c \in \mathbb{R}^+$ such that $Z(x) - c$ has fixed sign for $x < 0$,
- or for $x > 0$, the equation $Z(x_1) = Z(x_2)$ only has solutions for $0 < x_1 < \bar{x}_1 < x_g^+$, $x_g^+ < \bar{x}_2 < x_2$, with $\bar{x}_1 = x_0^+$, $\bar{x}_2 = x_\infty^+$ with $Z(x_\infty^+) = Z_\infty$. On the intervals $0 < x < \bar{x}_1$, $\bar{x}_2 < x$, $Z'(x) > 0$ holds true.

Case (6). For the function $Z(x)$ as defined in (4.9) under the strong focus condition (4.14) $\exists c \in \mathbb{R}$ such that $Z(x) - c$ has fixed sign for $x < 0$ or $x > 0$.

Case (7). For the function $Z(x)$ as defined in (4.9) under the strong focus condition (4.14)

- either for $x < 0$ the equation $Z(x_1) = Z(x_2)$ only has solutions for $x_1 < \bar{x}_1 < x_g^-$, $x_g^- < \bar{x}_2 < x_2 < 0$, with $\bar{x}_1 = x_0^-$, $Z(\bar{x}_2) = Z_\infty$. On the intervals $x < \bar{x}_1$, $\bar{x}_2 < x < 0$, $Z'(x) < 0$ holds true.
- or for $x > 0$ the equation $Z(x_1) = Z(x_2)$ only has solutions for $0 < x_1 < \bar{x}_1 < x_g^+$, $x_g^+ < \bar{x}_2 < x_2$, with $Z(\bar{x}_1) = Z_\infty$, $Z(\bar{x}_2) = x_0^+$. On the intervals $0 < x < \bar{x}_1$, $\bar{x}_2 < x$, $Z'(x) > 0$ holds true.

Case (8) can be reduced to Case (5) in Lemma 4.25 according to Lemma 4.20.

In Figures 13, 14, 15, and 16, an overview is given of all configurations.

Proof. The proof for the Cases (5), (6), (7), and (8) follows exactly the arguments of the proof for the cases (1), (2), (3), and (4) and is therefore omitted. $\square$

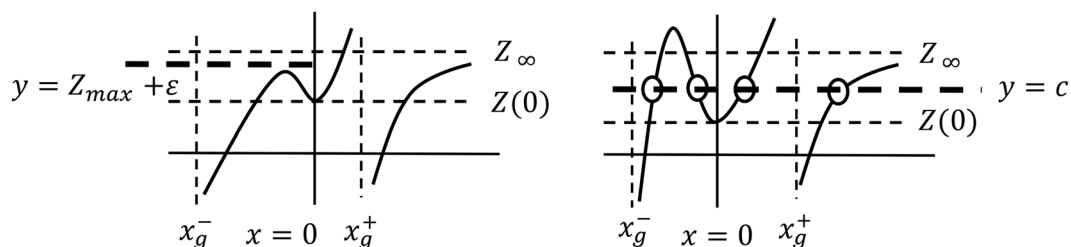


Figure 13: The graph of $y = Z(x)$ for case (5) of Lemma 4.25. There are two possibilities. Either there is a gap for $x < 0$, $y = Z_{\max} + \varepsilon$, or there are four intersections with $y = c$, $\forall c \in (Z(0), Z_\infty)$.

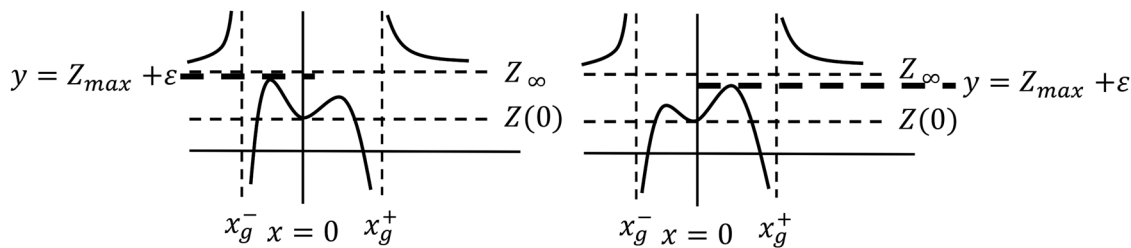


Figure 14: The graph of $y = Z(x)$ for case (6) of Lemma 4.25. There are two possibilities. Either there is a gap for $x < 0$, $y = Z_{\max} + \varepsilon$ or there is a gap for $x > 0$, $y = Z_{\max} + \varepsilon$.

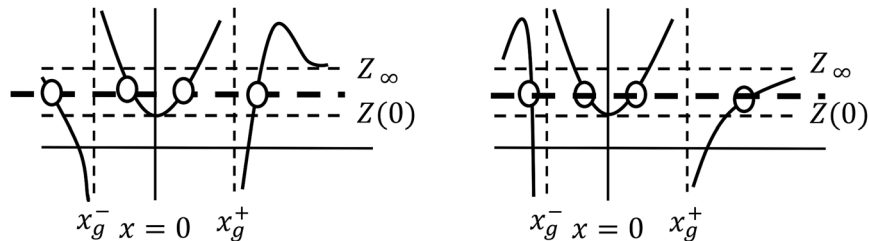


Figure 15: The graph of $y = Z(x)$ for case (7) of Lemma 4.25. There are four intersections with $y = c$, $\forall c \in (Z(0), Z_{\infty})$. This implies uniqueness of limit cycles for either $x < 0$ or for $x > 0$.

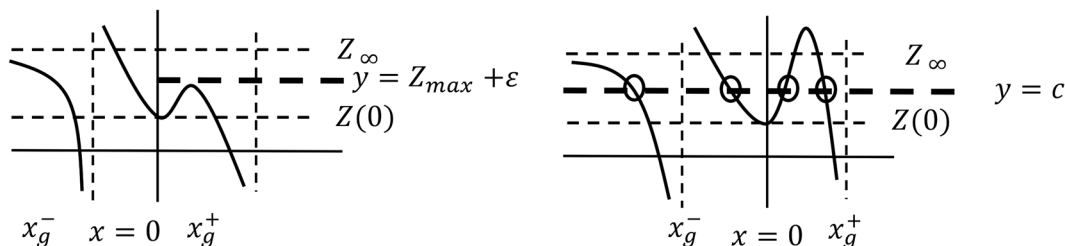


Figure 16: The graph of $y = Z(x)$ for case (8) of Lemma 4.25. There are two possibilities. Either there is a gap for $x > 0$, $y = Z_{\max} + \varepsilon$, or there are four intersections with $y = c$, $\forall c \in (Z(0), Z_{\infty})$.

Lemmas 4.21–4.25 essentially show for each case that either condition (iii) of 3.5 or (iii) of Theorem 3.6 is satisfied for $x < 0$ or $x > 0$. They form the basis of proving the distribution property for quadratic systems with two finite strong foci and a pair of complex singularities and an elementary saddle at infinity, and no invariant line.

We first translate all the lemmas into a proposition describing the number of limit cycles in each case.

Proposition 4.26. *The cases in Lemma 4.19 correspond to the following limit cycle distributions for system (3.1) with functions (4.6) and (4.7) under the restrictions of Lemmas 4.7 and 4.9 and the restriction that no singularity is weak (i.e., only strong foci are considered as in Definition 4.14). Here, the notation (m, n) with $m \in \{0\} \cup \mathbb{N}$, $n \in \{0\} \cup \mathbb{N}$, means m (n) limit cycles in the region $x < 0$ ($x > 0$).*

- **Case (1)** $(0, n)$ or $(m, 1)$.
- **Case (2)** $(m, 0)$ or $(0, n)$.
- **Case (3)** $(m, 1)$ or $(1, n)$.
- **Case (4)** $(m, 0)$ or $(1, n)$.
- **Case (5)** $(0, n)$ or $(m, 1)$.
- **Case (6)** $(m, 0)$ or $(0, n)$.
- **Case (7)** $(m, 1)$ or $(1, n)$.
- **Case (8)** $(m, 0)$ or $(1, n)$.

Proof. For each case, we apply either Theorems 3.5 or 3.6. For the two regions that we are interested in, $x < 0$ and $x > 0$, we take $r_1 = -\infty$, $r_2 = 0$, and $r_1 = 0$, $r_2 = \infty$, respectively.

In all cases, we note that the first two conditions of the theorems are satisfied, because in each region $x < 0$ or $x > 0$, we have exactly one zero of $g(x)$ and $f(x)$.

In both theorems, the function $F_{x_g}(x)$ is shifted by a constant K . For each region, constants are chosen in such a way that $F_{x_g}(x) + K$ becomes $F_0(x)$ given in (4.7).

The lemmas for each case then show the validity of condition (iii) of 3.5 or (iii) of Theorem 3.6. In the cases where we apply Theorem 3.5, the non-existence of limit cycle immediately follows, because each lemma contains an explicit construction of the constant c such that $Z(x) = c$ does not have a solution. This immediately applies to the cases where there is a gap in the graph of the function $y = Z(x)$.

For the cases with a unique limit cycle, in condition (iii) of Theorem 3.6, we take $F_{x_g}(x) + K = F_0(x)$. It remains to show that in the cases where we want to prove uniqueness of limit cycle that condition (iv) of Theorem 3.6 is satisfied.

The function $\frac{f(x)F_{x_g}(x)}{g(x)}$ can be rewritten in terms of $Z(x)$, using the fact that $Z(x) = \frac{f(x)F_0(x)}{g(x)}$, from which we deduce that $\frac{f(x)}{g(x)} = \frac{Z(x)}{F_0(x)}$. Substitution gives $\frac{f(x)F_{x_g}(x)}{g(x)} = \frac{Z(x)F_{x_g}(x)}{F_0(x)}$. The sign of the derivative of this function (which is needed in condition (iv) of Theorem 3.6) is then determined by $f(x)Z(x)K + Z'(x)F(x)F_0(x)$. For each of the factors in this expression, the sign is known (depending on the case we are considering). The only term for which the sign is not obvious a priori is the sign of K . The four cases where we need to check the sign of the derivative are Case (1), Case (3), (5), and (7). Cases (4) and (8) reduce to Cases (1) and (5), respectively, as stated earlier in Lemma 4.20. Since the proof for all cases is similar, we will only work out the proof for the first case in detail.

Case (1). Here for $x > 0$, we have $Z(x) > 0$, $Z'(x) < 0$, and $f(x) > 0$ ($f(x) > 0$), for $0 < x < \bar{x}_1 < x_g^+$ ($x_g^+ < \bar{x}_2 < x$). Also $F_0(x) < 0$, because neither $f(x)$ or $F_0(x)$ can have zeros for the intervals x for which $Z(x) > 0$. In particular, we note from the proof of Lemma 4.21 that $x_f^+ < x_g^+$. Any zeros of $F_0(x)$ can only appear for $\bar{x}_1 < x < x_g^+$. It follows that $F_0(x) < 0$ for all x in the intervals, $0 < x < \bar{x}_1 < x_g^+$ and $x_g^+ < \bar{x}_2 < x$. To find K , we observe that $F_{x_g}(x_g^+) + K = 0 + K = F_0(x_g^+) < 0$, i.e., $K < 0$. Moreover, $F_{x_g}(x) < 0$ on $0 < x < \bar{x}_1 < x_g^+$ and $x_g^+ < \bar{x}_2 < x$. This last statement is based on the fact that in the equation $F_{x_g}(x_1) = F_{x_g}(x_2)$ we need $x_2 > x_g^+$ according to the conditions of the uniqueness theorem, but on that interval, $F_{x_g}(x) < 0$. The situation is shown in Figure 17.

In the expression $f(x)Z(x)K + Z'(x)F(x)F_0(x)$, we know that $f(x) > 0$, for $0 < x < \bar{x}_1 < x_g^+$, so by choosing the interval $0 < x < \bar{x}_1 < x_g^+$, we obtain $f(x) > 0$, $Z(x) > 0$, $K < 0$, which shows that $f(x)Z(x)K < 0$. In the second term, $Z'(x) < 0$, $F(x) < 0$, $F_0(x) < 0$, which shows that $Z'(x)F(x)F_0(x) < 0$, and we conclude that $(\frac{fF_{x_g}}{g})'(x) < 0$ for $0 < x < \bar{x}_1$ as required by the first option (a) in condition (iv). It follows that all conditions of Theorem 3.6 are satisfied and that in the region $x > 0$ at most one limit cycle can occur.

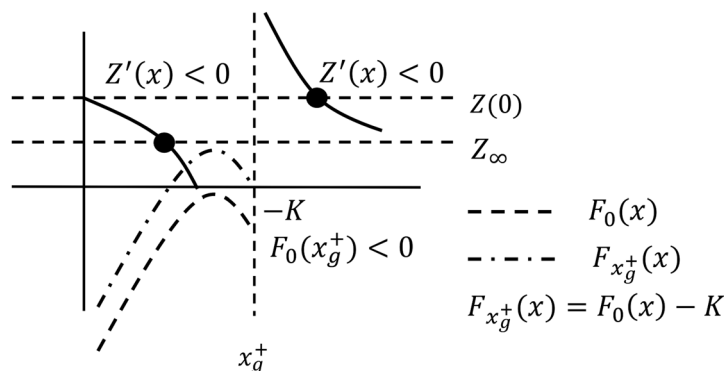


Figure 17: Illustration of the signs of $F_0(x)$, $F(x) < 0$ and K for Case (1) in the proof of Proposition 4.26.

For the other Cases (3), (5), and (7), the same arguments show that condition (iv) is satisfied, either option (a) or option (b) depending on the case. It should be noted that the condition that $F(r_2) \leq F(r_1)$ or $F(r_1) \leq F(r_2)$ is satisfied, because in all cases, the function $F_{x_g}(x) = F_0(x) - K$ will tend to $-\infty$ near the boundaries $x = r_1$ and $x = r_2$, i.e., we can apply the condition with the equality sign. $\square$

Finally, the main result of this section can be concluded using this proposition.

Theorem 4.27. *A quadratic system with two strong foci and two complex singularities and (at least) one elementary saddle at infinity, and no invariant line, can only have an $(n, 1)$ or $(n, 0)$ distribution of limit cycles surrounding the two foci where $n \in \{0\} \cup \mathbb{N}$.*

Proof. The quadratic system of this type was shown to belong to the family (4.1) in the Cherkas classification of quadratic systems. The condition $0 < Q_0 < 1$ in Lemma 4.7 implies that an elementary saddle exists at infinity. After transforming (4.1) to Liénard form (3.1) with functions (4.6) and (4.7), we showed that for at least one of the two foci always either Theorem 3.5 (non-existence of limit cycles) or Theorem 3.6 (uniqueness of limit cycles) holds true as indicated in Propositions 4.15 and 4.26 for the Cases (A) and (B) of Definition 4.13, proving the distribution property stated in the theorem. $\square$

5 The case of four real singularities

In the study by Zegeling and Kooij [17], a proof was given for this case which seems to be less complicated than the proof presented by Zhang [22]. It turns out that the proof of [17] can be simplified further.

In this case, there are four real finite singularities, two of which are foci, the other two we label as A_3 and A_4 respectively. After an affine transformation, we place A_3 and A_4 on the y -axis, one of them at the origin and the other at $(0, 1)$:

$$\begin{aligned}\frac{dx}{dt} &= P_2(x, y) = a_{10}x - a_{02}y + a_{20}x^2 + a_{11}xy + a_{02}y^2, \\ \frac{dy}{dt} &= Q_2(x, y) = b_{10}x - b_{02}y + b_{20}x^2 + b_{11}xy + b_{02}y^2.\end{aligned}\tag{5.1}$$

This system allows for a surprisingly easy transformation to a Liénard system. After the transformation (for more details about the origin of this transformation, see the study by Zegeling and Kooij [17]) $v = x$, $u = y/x$, followed by $x = L(u)e^z$, where $\frac{L'(x)}{L(x)} = -\frac{g_2(x)}{f_1(x)}$ system (5.1) becomes the generalized Liénard system (and writing $u \rightarrow x$ and $z \rightarrow y$):

$$\begin{aligned}\frac{dx}{dt} &= F(x) - e^y, \\ \frac{dy}{dt} &= g(x),\end{aligned}\tag{5.2}$$

with

$$\begin{aligned}F(x) &= \frac{f_0(x)}{f_1(x)L(x)}, \\ g(x) &= -\frac{g_1(x)f_1(x) + g_2(x)f_0(x)}{f_1^2(x)L(x)}, \\ f(x) = F'(x) &= \frac{f'(x)f_1(x) - f_0(x)f_1'(x) + g_2(x)f_0(x)}{f_1^2(x)L(x)},\end{aligned}\tag{5.3}$$

and the notation $f_0(x) = b_{10} - (b_{02} + a_{10})x + a_{02}x^2$, $f_1(x) = -b_{20} + (a_{20} - b_{11})x + (a_{11} - b_{02}x^2 + a_{02}x^3$, $g_1(x) = a_{10} - a_{02}x$, $g_2(x) = a_{20} + a_{11}x + a_{02}x^2$.

This is a special case of the generalized Liénard system

$$\begin{aligned}\frac{dx}{dt} &= F(x) - \psi(y), \\ \frac{dy}{dt} &= g(x),\end{aligned}\tag{5.4}$$

with $\psi(y) = e^y$.

This transformation is particularly useful, because the function $\frac{f(x)}{g(x)}$, which is used in the so-called Zhang Zhifen theorem, simplifies to a simple rational function:

$$\frac{f(x)}{g(x)} = \frac{c_3x^3 + c_2x^2 + c_1x + c_0}{d_2x^2 + d_1x + d_0},\tag{5.5}$$

where the constants c_i and d_i depend on the coefficients a_{ij} , b_{ij} in (5.1). In the following, we will not need the explicit form of these coefficients. The function (5.5) is a rational function with a cubic function in the numerator and a quadratic function in the denominator. The quadratic function representing the zeros of $g(x)$ has exactly two zeros by construction, because they correspond to the two foci of the original quadratic system. Since the transformation $u = y/x$ removed the singularities at $x = 0$, only two singularities of the original four remain: the two foci around which limit cycles can form two nests.

To prove the distribution property in Definition 1.1 of limit cycles for this class, two theorems similar to Theorems 3.5 and 3.6 will be used with two differences. Firstly, the Liénard system (5.2) contains a term e^y , i.e., a generalized Liénard system where Theorem 3.6 does not apply (even though some extensions do exist). Fortunately, for system (5.2), we will not need this strong theorem: the classical Zhang-Zhifen theorem extended to generalized Liénard systems will be sufficient. Explicitly it states:

Theorem 5.1. (Uniqueness) [19,20] *Suppose system (5.4) with differentiable $F(x) = F_{x_g}(x) = \int_{x_g}^x f(\bar{x})d\bar{x}$, and continuous $g(x)$, $f(x)$, satisfies the following conditions on the interval $r_1 < x < r_2$.*

- (i) $\frac{d\psi(y)}{dy} > 0$,
- (ii) $x_g \in (r_1, r_2)$ exists such that $(x - x_g)g(x) > 0$, for $x \neq x_g$, and $g(x_g) = 0$,
- (iii) $\frac{f(x)}{g(x)}$ is nondecreasing (or nonincreasing) for all $x \in (r_1, r_2)$.

Then (5.4) has at most one limit cycle in the strip $r_1 < x < r_2$, and if it exists, it is hyperbolic.

For proving the non-existence of limit cycles, we will use the natural extension of Theorem (3.5):

Theorem 5.2. (Non-existence adjusted) [19] *Suppose system (3.1) with differentiable $F(x) = F_{x_g}(x) = \int_{x_g}^x f(\bar{x})d\bar{x}$, and continuous $g(x)$, $f(x)$, satisfies the following conditions on the interval $r_1 < x < r_2$.*

- (i) $\frac{d\psi(y)}{dy} > 0$,
- (ii) $x_g \in (r_1, r_2)$ exists such that $(x - x_g)g(x) > 0$, for $x \neq x_g$, and $g(x_g) = 0$,
- (iii) $x_f \in (r_1, r_2)$ exists such that $(x - x_f)f(x) < 0$ for $x \neq x_f$, and $f(x_f) = 0$, $x_f \neq x_g$,
- (iv) $\exists c \in \mathbb{R}$ such that $f(x) - cg(x)$ has fixed sign in $r_1 < x < r_2$.

Then (5.4) does not have limit cycles in the strip $r_1 < x < r_2$.

The proof of this theorem goes along similar lines as for Theorem 3.5 showing that the system of equations $\frac{f}{g}(x_1) = \frac{f}{g}(x_2)$, $F(x_1) = F(x_2)$ cannot have a solution since the first equation cannot have a solution if $f(x) - cg(x)$ has fixed sign.

To apply these two theorems to system (5.2), we need a simple fact about limit cycles surrounding different singularities.

Lemma 5.3. *Suppose two limit cycles surround two different foci in system (5.2) with $\frac{d\psi(y)}{dy} > 0$, then any vertical line $x = c$ can only be crossed by one of the two limit cycles.*

Proof. On each line $x = c$, we have $\frac{dx}{dt} = F(c) - \psi(y)$. Since $\frac{d\psi(y)}{dy} > 0$, there is at most one value $y = y^*(c)$ = $\psi^{-1}[F(c)]$, where $\frac{dx}{dt} = 0$. Moreover, $\frac{dx}{dt} > 0$ ($\frac{dx}{dt} < 0$) for $y < y^*(c)$ ($y > y^*(c)$). It means that two limit cycles crossing the line $x = c$ would lie in the same nest which contradicts the assumption on the limit cycles. Therefore, only one of the two limit cycles can cross $x = c$. $\square$

Lemma 5.3 provides the necessary tool for a simple proof of the distribution property of system (5.1).

Theorem 5.4. *A quadratic system with four real singularities has a $(1, 1)$ or $(n, 0)$ distribution of limit cycles surrounding the two foci where $n \in \{0\} \cup \mathbb{N}$.*

Proof. After transformation into a Liénard system (5.4) with functions given by (5.2), the function $\frac{f(x)}{g(x)}$ is a rational function with cubic numerator and quadratic denominator. Consider the intersection of the graph of $y = \frac{f(x)}{g(x)}$ with a horizontal line $y = c$. The number of intersections is given by the number of solutions to the cubic equation $f(x) - cg(x)$.

Since the equation is cubic we have the following possibilities. For every $c \in \mathbb{R}$ there are exactly three solutions. In that case, the derivative of the function $\frac{f(x)}{g(x)}$ is never zero, and the function is monotonically decreasing (or increasing) for all x . Since $\frac{d\psi(y)}{dy} = e^y > 0$, according to Theorem 5.1, at most one limit cycle occurs around each focus, i.e., a $(1, 1)$, $(1, 0)$, or $(0, 0)$ distribution.

Suppose next that there is (at least) one c for which $f(x) - cg(x)$ has less than three zeros. In the case of two zeros, one of the zeros has to be a double zero, and we can perturb c in such a way that only 1 zero exists. If there is exactly one zero x^* for some c^* , according to Lemma 5.3, only limit cycles surrounding one of the two foci can intersect the line $x = x^*$. It means that the limit cycles surrounding the other focus do not cross it and lie in the region to one side of the vertical line $x = x^*$. Since in that region necessarily $f(x) - c^*g(x)$ has fixed sign, and no limit cycles surround the focus according to Theorem (5.2), the only possible distribution is of the type $(n, 0)$.

Finally, if $f(x) - cg(x)$ does not have zeros for some c (i.e., if the numerator reduces to a quadratic function), then no limit cycles exist surrounding any focus, i.e., a $(0, 0)$ distribution. $\square$

Remark 5.5. The main reason why the proof in this case is so much simpler than the case with two finite foci and a pair of complex singularities is that the transformation to a Liénard system removed two singularities and the ensuing function $g(x)$ was relatively simple. We have tried to apply the same idea to the case of two finite foci and a pair of complex singularities, but were not able to achieve any progress. The proof for that case would probably be much simpler if such a transformation could be found.

6 Non-generic cases

To complete the proof of the distribution property, we need to cover the remaining cases which are non-generic in the sense that they can be perturbed into one of the cases which we have already covered.

The cases remaining are the following according to Lemma 2.3.

- (1) (Generic) proved in Theorem 4.27 except for the case of a weak focus and the case of a higher order saddle at infinity. The case of an invariant line was proved in Lemma 4.2,
- (2) (Generic) proved in Theorem 5.4,
- (3) (Non-generic) to be proved.

Here, we view the three subsets of (3) corresponding to $p_1^2 - 4p_2 = 0$ in (2.4), $p_1 \neq 0, p_2 = 0$ in (2.4) and $p_1 = 0, p_2 = 0$ in (2.4), respectively, as sets which can be perturbed into the case (2) (or (1)) by setting $p_1^2 - 4p_2 = \varepsilon$, $p_2 = \varepsilon$ and $p_2 = \varepsilon$ respectively with $0 < \varepsilon \ll 1$.

For the two remaining cases in subset (1), we will use different canonical forms. For the case of a higher order saddle, we will use system (2.4), while for the case of a weak focus, we will use class III of the Chinese classification (2.1). We note that in the book [1] in section (2.2), the method of perturbing a non-generic case was already mentioned, and it was observed that “either we have a double finite singularity or a multiple infinite singularity coming from a coalescence of finite and infinite singularities, and by means of a perturbation, ...”.

To deal with the remaining non-generic cases, we will use the concept of rotated vector fields as explained in the following section.

6.1 Rotated vector field parameters

Consider a general planar autonomous system depending on a parameter $\alpha \in \mathbb{R}$

$$\begin{aligned}\frac{dx}{dt} &= P(x, y, \alpha), \\ \frac{dy}{dt} &= Q(x, y, \alpha),\end{aligned}\tag{6.1}$$

where $P(x, y, \alpha), Q(x, y, \alpha)$, are analytical functions of x, y, α . See the books presented by Ye et al. and Zhang et al. [14,19] for the details and proofs of the following standard lemmas about rotated vector fields.

Lemma 6.1. *If*

$$\left| \frac{P(x, y, \alpha)}{dP(x, y, \alpha)/d\alpha} - \frac{Q(x, y, \alpha)}{dQ(x, y, \alpha)/d\alpha} \right| > 0,\tag{6.2}$$

for $\alpha = \alpha_0 \in \mathbb{R}$, $(x, y) \in \Omega$, where Ω is some open region in the phase plane, then the vector field defined by $(P(x, y, \alpha), Q(x, y, \alpha))$ rotates counter clockwise (clockwise) for $\alpha = \alpha_0 + \varepsilon$, with $\varepsilon > 0$ ($\varepsilon < 0$), where $0 < |\varepsilon| \ll 1$.

If the determinant (6.2) is negative, then the vector field rotates in the opposition direction.

The parameter α is referred to as a rotated vector field parameter.

Lemma 6.2. *If (6.1) has a rotated vector field parameter as defined in Lemma 6.1 satisfying 6.2, then a counter clockwise semistable limit cycle L_{su} that is stable from the inside will break into two limit cycles L_s and L_u for $\alpha = \alpha_0 + \varepsilon$, with $\varepsilon > 0$, where $0 < |\varepsilon| \ll 1$. L_s (L_u) is a stable (unstable) limit cycle lying inside (outside) L_{su} . For $\varepsilon < 0$, the semistable limit cycle will disappear. If L_{su} is unstable from the inside, it will disappear for $\varepsilon > 0$, and it will break into two limit cycles L_u and L_s with L_u (L_s) an unstable (stable) limit cycle.*

In practice, the lemma is used informally by saying: every semi-stable limit cycle can be split by a rotational parameter if the direction of movement of the parameter is chosen correctly. The lemma is still true if the vector field rotates on all points of the limit cycle, except perhaps for a finite number where the vector field does not change direction (i.e., a finite number of points where (6.2) is equal to zero).

These lemmas can be applied to system (2.4). In particular, we can identify parameters rotating the vector field in that system.

Lemma 6.3. *In system (2.4):*

- the parameter b rotates the vector field in both regions $x < 0$ and $x > 0$ in the same direction if we define $a_{11} = a + b, b_{11} = b$,
- the parameter p_1 rotates the vector field in the regions $x < 0$ and $x > 0$ in opposite directions.

Proof. The determinant (6.2) of Lemma 6.1 becomes ax^2y^2 for the parameter b and axy^2 for the parameter p_1 proving the statement of the lemma. $\square$

We note that in system (2.4) more rotational parameters can be detected using the fact that $y = x$ is a line without contact, i.e., a line that cannot be crossed by limit cycles. However, for the purposes of this article, they will not be needed. The combination of two rotated vector field parameters allows for an independent rotation direction in each region $x < 0$ and $x > 0$. This method was used before by [4] in the study of bifurcation surfaces in the parameter space of quadratic systems.

We can apply these rotational parameters to show that some of the non-generic cases satisfy the distribution property.

For the case of a weak focus in subset (1) of Lemma 2.3, we will use class III of the Chinese classification (2.1).

Lemma 6.4. *In system (2.1):*

- the parameter δ rotates the vector field for all x in the same direction,
- the parameter s with $\delta = s$, $l = as$, $m = bs$ rotates the vector field in the regions $1 + ax + by > 0$ and $1 + ax + by < 0$ in opposite directions.

The second rotational parameter s is rotating in different directions in the half planes defined by $1 + ax + by = 0$. This line is a line which cannot be crossed by limit cycles surrounding the singularity at $(0, 0)$ and at $(1, \frac{1}{n})$, i.e., it separates the two nests of limit cycles in this class after we have positioned the two foci at $(0, 0)$ and $(0, \frac{1}{n})$.

The conclusion is that for systems (2.1) and (2.4), we can find two rotational parameters which can rotate the vector field in independent directions in both regions where the nests of limit cycles occur. It means that no matter what kind of semistable limit cycles we have in both nests, we can always perturb both of them by choosing the rotational parameters in an appropriate way such that (at least) two limit cycles in each nest are created. This would obviously contradict the distribution property for the generic cases.

6.2 The perturbation of non-generic cases

The idea behind proving the distribution property for the remaining non-generic cases is that a small perturbation will generate a generic case for which we have already proved the distribution property. One way to proceed would be to try and apply the non-existence Theorem 3.5 and uniqueness Theorem 3.6 directly to the non-generic systems. This was the strategy followed in [22]. This leads to complicated calculations which in our opinion are not necessary, because of a perturbation argument.

If a non-generic case does not have the distribution property, then necessarily there must be a system for which there is a nest with an (n, m) distribution with $n \geq 2, m \geq 2$. For each limit cycle which is not semi-stable, a perturbation into a generic case will generate (at least) one limit cycle. Therefore, the only case for which the existence of an (n, m) distribution with $n \geq 2, m \geq 2$ in the non-generic case could be possible, is when all, except possibly one limit cycle, in a nest in the non-generic case are semi-stable limit cycles which disappear under perturbation.

However, if we perturb with a rotational vector field parameter, we can always ensure that there is a perturbation direction where the semistable limit cycle would not disappear, but would break into two limit cycles. This would give a contradiction with the distribution property of the generic case.

Another way would be (if we cannot find a perturbation parameter which rotates the vector field) is to find a rotational vector field parameter within the class of non-generic systems and use it to break the semi-stable limit cycle into a stable and unstable limit cycle. Then perturbing into the generic case would give a contradiction with the distribution property as well. As shown in the previous section, for both canonical

forms (2.1) and (2.4), we can always find a change of rotational parameters such that semistable limit cycles in two nests would break into (at least) four limit cycles.

Proposition 6.5. *A quadratic system with two foci of which at least one is a weak focus and two complex singularities and (at least) one elementary saddle at infinity can only have an $(n, 1)$ or $(n, 0)$ distribution of limit cycles surrounding the two foci where $n \in \{0\} \cup \mathbb{N}$.*

Proof. The proof is by contradiction. Since the system must have a weak focus, we can assume that it is positioned in class III (2.1) at the origin with $\delta = 0$. Suppose there is such a system with an (n, m) distribution with $n \geq 2, m \geq 2$. If there are two or more limit cycles in each nest which are not semi-stable, then with $\delta = \varepsilon$, $0 < \varepsilon \ll 1$ the origin becomes a strong focus. Choosing $s \neq 0$, the other focus at $(0, \frac{1}{n})$ will become strong as well (if it were weak too to begin with). The resulting system would have two strong foci for which the distribution property was proved in Theorem 4.27, i.e., a contradiction occurs. Therefore, in one or both of the nests, a semistable limit cycle must occur. However, by choosing a suitable rotation with the parameters δ and s from Lemma 6.4, we can break the semi-stable limit cycles and obtain an (n, m) distribution with $n \geq 2, m \geq 2$ for the case with two strong foci, again contradicting Theorem 4.27.

Hence, the original assumption was wrong and the case with a weak focus needs to satisfy the distribution property as given in Definition 1.1. $\square$

Proposition 6.6. *A quadratic system with two strong foci and two complex singularities and (at least) one higher order saddle at infinity can only have an $(n, 1)$ or $(n, 0)$ distribution of limit cycles surrounding the two foci where $n \in \{0\} \cup \mathbb{N}$.*

Proof. The proof is by contradiction. It follows the same argument as for the previous lemma, except that now we do not have a perturbation necessarily creating a generic case. By changing rotational parameters b_{11} and p_1 any system (2.4) with an (n, m) distribution with $n \geq 2, m \geq 2$ can be broken into a system with at least (n, m) limit cycles which are not semi-stable. Perturbing that system into the case with an elementary saddle at infinity would lead to a contradiction with Theorem 4.27.

Hence, the original assumption was wrong, and the case of two finite foci with a higher order saddle at infinity needs to satisfy the distribution property as given in Definition 1.1. $\square$

Proposition 6.7. *A quadratic system with a two foci and either no other finite singularities, or a finite real higher order singularity, or one elementary finite singularity, can only have an $(n, 1)$ or $(n, 0)$ distribution of limit cycles surrounding the two foci where $n \in \{0\} \cup \mathbb{N}$.*

Proof. The proof is exactly the same as in the previous lemma. By changing rotational parameters b_{11} and p_1 any system (2.4) with an (n, m) distribution with $n \geq 2, m \geq 2$ can be broken into a system with at least (n, m) limit cycles which are not semi-stable. Perturbing that system into the case with four singularities (in the case of three elementary singularities this would require taking $p_2 = \varepsilon$) would lead to a contradiction with Theorem 5.4. In the case of when there are no other finite singularities, first, a perturbation could be made with $p_1 = \varepsilon$ bringing it into one of the other cases of the proposition.

Hence, the original assumption was wrong, and in all three cases, the distribution property as given in Definition 1.1 must hold. $\square$

A consequence of the previous three propositions is that in a quadratic system with a weak focus of order two or three (three being the highest possible order in quadratic systems), the limit cycle surrounding another focus is unique. It is well known that more than one limit cycle can be perturbed from such a higher order weak focus within the class of quadratic systems. It also implies that in the original examples of the $(n \geq 3, m \geq 1)$ distribution by [2,11] the perturbed system must have a distribution of exactly $(3, 1)$ limit cycles, because in the study by Li [10], it was proved that in a quadratic system a weak focus of order three is not surrounded by other limit cycles.

Corollary 6.8. *In a quadratic system with two foci of which at least one is a weak focus at most one limit cycle surrounds another focus if the order of the weak focus is two or three.*

The aforementioned lemmas cover the remaining non-generic cases of the subsets (1) and (3) in Lemma 2.3.

7 Main result

In the previous sections for several cases in quadratic systems (1.1), the distribution property as given in Definition 1.1 was proved. We collect them to obtain the main result of this article Theorem 1.2.

Proof. In Lemma 2.3, three families of (1.1) were identified where a distribution of limit cycles could occur. For each case, we proved the distribution property according to:

- (1) (Generic) There are exactly two foci and one pair of finite complex singularities $\Rightarrow$ Theorem 4.27, Proposition 6.5, Proposition 6.6, Lemma 4.2.
- (2) (Generic) There are exactly four real finite singularities of which two are foci $\Rightarrow$ Theorem 5.4.
- (3) (Non-generic)
 - There are exactly three real finite singularities of which two are foci and the third a singularity with eigenvalue equal to zero $\Rightarrow$ Proposition 6.7.
 - There are exactly three real finite singularities of which two are foci and the third an elementary saddle $\Rightarrow$ Proposition 6.7.
 - There are exactly two foci and no other finite singularities $\Rightarrow$ Proposition 6.7. □

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References

- [1] J. C. Artés, J. Llibre, D. Schlomiuk, and N. Vulpe, *Geometric Configurations of Singularities of Planar Polynomial Differential Systems*, Birkhäuser, 2021, 711 pages.
- [2] L. Chen and M. S. Wang, *Relative position and number of limit cycles of a quadratic differential system*, Acta Math. Sinica **22** (1979), 751–758. (Chinese)
- [3] L. A. Cherkas, *Cycles of the equation $y' = Q_2(x, y)/P_2(x, y)$* , Differentsial'nye Uravneniya **9** (1973), 1432–1437. (Russian), Differential Equations **9** (1973), 1099–1103.
- [4] L. A. Cherkas, *Bifurcation of limit cycles of a quadratic system with variation of the parameter rotating the field*, Differentsial'nye Uravneniya **17** (1981), 2002–2016 (Russian), Differential Equations **17** (1981), 1265–1276.
- [5] B. Coll and J. Llibre, *Limit cycles for a quadratic system with an invariant straight line and some evolution of phase portraits*, Coll. Math. Soc. J. Bolyai **53** (1988), 111–123, Qual. Theory of Diff. Eq.
- [6] W. A. Coppel, *Some quadratic systems with at most one limit cycle*, Dyn. Rep. Ser. Dyn. Sys. Appl. **2** (1989), 61–68.
- [7] W. A. Coppel, *A survey of quadratic systems*, J. Diff. Equ. **2** (1966), 293–304.
- [8] J. Dong, *Positions of limit cycles of the system $dx/dt = \sum_{0 \leq i+k \leq 2} a_{ik}x^i y^k$, $dy/dt = \sum_{0 \leq i+k \leq 2} b_{ik}x^i y^k$* , Sci. Sinica **8** (1959), 151–171.
- [9] F. Dumortier and C. Rousseau, *Cubic Liénard equations with linear damping*, Nonlinearity **3** (1990), 1015–1039.

- [10] C. Li, *Non-existence of limit cycle around a weak focus of order three for any quadratics system*, Chinese Ann. Math. Ser. B **7** (1986), no. 2, 174–190.
- [11] S. Shi, *A concrete example of the existence of four limit cycles for plane quadratic systems*, Sci. Sinica **23** (1980), 153–158, (Chinese).
- [12] L. V. Sakhova, *Straight isoclines*, Trud. Samar. Gos. Univ. (N.S.) Vyp. **144** (1964), 93–105, (Russian).
- [13] A. P. Vorob'ev, *Cycles about a singular point of the node type*, Dokl. Akad. Nauk BSSR (1960), 369–371 (Russian).
- [14] Y. Ye, S. Cai, L. Chen, K. Huang, D. Luo, Z. Ma, et al., *Theory of limit cycles*, The American Mathematical Society, Providence, Rhode Island, 1984, 435 pages.
- [15] Y. Ye, *Qualitative Theory of Polynomial Differential Systems*, Shanghai Science-Technical Publisher, Shanghai, 1995, (Chinese).
- [16] S. Yu, *Limit cycles of quadratic systems*, Acta Math. Sinica **20** (1977), 193–205 (Chinese).
- [17] A. Zegeling and R. E. Kooij, *The distribution of limit cycles in quadratic systems with four finite singularities*, J. Diff. Equ. **151** (1999), 373–385.
- [18] A. Zegeling, *Limit Cycles in Quadratic Systems*, PhD thesis Technical University Delft, 1991, p. 157.
- [19] Z. Zhang, T. Ding, W. Huang, and Z. Dong, *Qualitative theory of differential equations*, The American Mathematical Society, Providence, Rhode Island, 1992, 461 pages.
- [20] Z. Zhang, *Proof of the uniqueness theorem of limit cycles of generalized Liénard equations*, Appl. Anal. **23** (1986), 63–76.
- [21] Y. Zhou, C. Wang, and D. Blackmore, *The uniqueness of limit cycles for Liénard system*, J. Math. Anal. Appl. **304** (2005), no. 2, 473–489.
- [22] P. Zhang, *On the distribution and number of limit cycles for quadratic systems with two foci*, Qual. Theory Dyn. Sys. **3** (2002), 437–463.